FIREWORK CHRONOS · FOR CLAUDE CODE

Claude Code Build Spec

Everything needed to build the website exactly as designed: rules, design reference, memory setup, data sources, APIs, 46 formulas, page-by-page specs with images, API contract, database schema and milestones.



Dhruv Sharma and team · 20 October 2026

Free tiers change often — each limit carries the date it was checked.

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START HERE

1 How to use this document with Claude Code

This single document is the full build specification for the FIREWORK CHRONOS website, backend, data pipeline and AI features for NASA Space Apps 2026 (challenge: Harmonization of MODIS and VIIRS Hot Spots). It collects the research, data sources, APIs, formulas, page designs with images, API contracts, database schema, build order and rules from all earlier documents. Give it to Claude Code together with the files listed below, and Claude Code has everything except your keys and your data files.

1.1 Files to put in the repository before starting

FILE	WHERE TO PUT IT	WHY
This PDF (and CLAUDE_BUILD_SPEC.md)	docs/spec/	the specification Claude Code follows
CLAUDE.md (section 'Starter CLAUDE.md' below)	repository root	Claude Code reads it automatically in every session
FIREWORK_data_kit.zip (unzipped)	data_kit/	data_sources.json, fx_engine.py, make_data_index.py, sample API responses
SpaceApps_code.zip (unzipped)	pipeline/ and tests/	tested harmonisation and accuracy code
Home_hero_kit.zip (unzipped)	frontend/public/hero/	three.js Earth scene, textures
FIREWORK_design_reference.zip (unzipped)	docs/design/	EXACT look: HTML/CSS mock-ups of every page, tokens, fonts, screenshots
.env (you create it, never commit)	repository root	your API keys

1.2 What you must provide (Claude Code cannot do these)

1. API keys and accounts: NASA Earthdata, FIRMS MAP_KEY, Google Earth Engine noncommercial project, OpenAQ key, Supabase project URL and keys, Groq / OpenRouter / Gemini keys (see chapter 7).
2. The FIRMS archive download for India (MODIS 2000→, VIIRS 2012→), requested from the FIRMS download page and saved as CSV in data/raw/.
3. Earth Engine authentication on your machine (earthengine authenticate) — this needs your Google login.
4. Decisions marked OPEN in chapter 13 (2026 rules on pre-built code, final region list).
5. Review of every number before the demo: nothing synthetic may be shown as a result.

1.3 Prompt to paste into Claude Code

```
Read CLAUDE.md, docs/spec/CLAUDE_BUILD_SPEC.md (and the PDF in docs/spec/ for images) and docs/d
    esign/README.md.
Work milestone by milestone from chapter 12 (M0 to M8). Build the exact look first (M1) from doc
    s/design, then the data. For each milestone:
1. Restate the tasks and the 'done when' checks.
2. Implement them, following the rules in chapter 3 exactly. Pages must match docs/design/previe
    w/*.html and screenshots/*.jpg exactly.
3. Run the tests and show me the result.
4. Stop and wait for my OK before the next milestone.
Never invent data. If a key or data file is missing, stop and tell me which one.
```

Which Claude model to choose in Claude Code

Use the most capable Opus model your plan offers (switch with /model) for planning, the exact-look pages (M1), the pipeline and the science (M2–M4): these need careful reading of the spec and images. A Sonnet model is fine for routine follow-ups (small UI fixes, tests, docs) and uses less of your limit. Turn on extended thinking or plan mode for each milestone, and ask Claude Code to show a plan before editing.

Tip

Claude Code reads PDFs in parts (about 20 pages at a time) and sees the images. The Markdown copy (CLAUDE_BUILD_SPEC.md) has the same text and is faster for it to search, so keep both in docs/spec/.

BRIEF

2 What we are building and why

2.1 The project in one paragraph

FIREWORK CHRONOS turns 25 years of satellite fire detections from two instrument families — MODIS (Terra 2000→, Aqua 2002→) and VIIRS (Suomi NPP 2012→, NOAA-20 2018→, NOAA-21 2023→) — into one calibrated burning calendar on a single scale (VIIRS-equivalent fire cell-days), with an uncertainty band on every value, a label for the type of fire source, and a warning where satellites probably missed burning. Users can explore it on a globe, read a month-by-year calendar, check every formula, and upload their own data to compare.

2.2 Why now

FACT	DATE	SOURCE
Suomi NPP observations stop	2 Nov 2026	NASA Earthdata alert
Terra MODIS data collection ends	Jan 2027	CAMS / LAADS
Aqua MODIS ends	~Sep 2027	CAMS / LAADS
Challenge statements + datasets released	28 Oct 2026	Space Apps
Hackathon	14–15 Nov 2026	Space Apps
Judging starts	16 Nov 2026	Space Apps

2.3 Local story: Punjab and Haryana

iFOREST's 2025 report says more than 90% of large Punjab farm fires in 2024–25 happened after 3 PM (about 3% in 2021), and burnt area fell much more slowly (about 31,447 km^2 in 2022 to about 20,000 km^2 in 2025) than fire counts. Satellites pass at about 10:30 and 13:30 local time, so late burning is missed. A count-only calendar would wrongly say burning collapsed; our blind-spot flag (F45, F46) exists for this.

2.4 Judging criteria (equal weight) and how we answer them

CRITERION	OUR EVIDENCE	PAGE
Influence	Punjab–Haryana air-quality story, one-page briefs for officials	Home, Brief (L)
Creativity	blind-spot flag, source attribution, user data studio	I, H, E
Validity	tested formulas, leave-one-year-out, seam test, honest bands	G, F
Relevance	harmonised 2000→ calendar is the core ask	C

CRITERION	OUR EVIDENCE	PAGE
Presentation	2-minute demo, clean UI, realistic Home hero	A, video

2.5 Competitors (self-reported, unreviewed)

REPO	WHAT IT DOES	OUR GAP
FireCal	global 0.1° cell-days, per-area monthly ratio, LOYO validation	no industrial separation, no diurnal bias, no burned-area check
Kiln Watch	Bangladesh, cloud-normalised cell-days with 95% bands, seam test	one country, weak source classifier
Chronofire / TerraHarmonia	plans / early stage	not built

RULES

3 Rules Claude Code must follow

3.1 Code style (the team's preference)

- Beginner-readable code. React: only useState, useEffect, useParams, useNavigate, props, .map(), .filter(), .find() and if/else blocks.
- No custom hooks, no ternaries for major branching (use if/else before return), no comments in code, no heavy abstractions.
- No new npm/pip packages without asking first. The approved list is in chapter 6.
- Always output full files, not snippets.
- Python: plain functions, if/else, loops; type hints welcome; pandas/polars allowed.

3.2 Data honesty

- Never invent, hard-code or 'example' a number in the UI. Every number comes from the API.
- Every API response is {data, meta}. meta has data_source, version, generated_at, units, demo.
- If meta.demo is true the UI shows a DEMO chip. Synthetic numbers are never presented as results.
- If a source fails, show CACHED or 'unavailable' — never zero.
- The LLM may write sentences but never numbers; numbers are inserted from the API JSON.

3.3 Security and git

- Keys only in .env (local) and platform secrets. .env is in .gitignore. The browser never sees a key.
- Do not commit data/raw or large files; use Cloudflare R2 for Parquet.
- Never push to main without the user's OK; one branch per milestone.
- Validate every upload (size $\leq$ 25 MB, type allow-list); never execute uploaded content.

3.4 Starter CLAUDE.md

```
# FIREWORK CHRONOS
NASA Space Apps 2026 - Harmonization of MODIS and VIIRS Hot Spots.
Spec: docs/spec/CLAUDE_BUILD_SPEC.md (+ PDF with images).

## Stack
frontend/ React 19 + Vite, MapLibre GL, deck.gl, Recharts, d3, three.js (Home hero only)
backend/  FastAPI, SQLModel/SQLAlchemy + GeoAlchemy2, Supabase Postgres + PostGIS
pipeline/ Python: pandas/polars, duckdb, earthengine-api, earthaccess; Parquet on Cloudflare R2
data_kit/ data_sources.json + fx_engine.py (source fallback resolver)

## Rules
- Beginner-readable: React only useState/useEffect/useParams/useNavigate/props/.map/.filter/.fin
    d, if/else before return, no ternaries for branching, no comments, no custom hooks, no new p
    ackages without asking.
```

```
- Full files, not snippets.
- Every API response = {data, meta}; meta.data_source always set; meta.demo drives the DEMO chip
    .
- Never invent numbers. LLM writes words only.
- Keys only in .env / platform secrets. Never commit .env or data/raw.
- The look must match docs/design (HTML mock-ups + screenshots) exactly. Build the look first wi
    th fixture data (M1), then real data.
- Read graphify-out/GRAPH_REPORT.md and DATA_INDEX.json before opening raw files; never load ful
    l data into context.
- Work milestone by milestone (spec chapter 12); stop after each for review.

## Commands
backend:  uvicorn app.main:app --reload ; pytest
frontend: npm run dev ; npm run build
pipeline: python -m pipeline.run --region punjab_haryana
```

DESIGN

4 Exact look first: the design reference is the ground truth

The website must look exactly like the mock-ups in this document — same layout, spacing, colours, fonts, panels, chips and globe style. The mock-ups were built as real HTML/CSS pages, so nothing has to be guessed: FIREWORK_design_reference.zip contains the source of every page (preview/*.html), the shared stylesheet atlas.css with every token and component class, the fonts, and a screenshot of each page. Claude Code must open these files and rebuild each page in React with the same structure.

Build order follows your wish: the full website look comes first. In milestone M1 every page is built pixel-faithfully using fixture JSON (copied from the mock-up values and marked demo:true, so the DEMO chip shows). Real data replaces the fixtures later, page by page, without changing the look.

4.1 How Claude Code checks it matches

1. Serve docs/design (python3 -m http.server 8765) and open preview/<page>.html.
2. Build the React page with the same class names from atlas.css (copy atlas.css into frontend/src/styles/).
3. Take a Playwright screenshot of the React page at 1600 × 1000 and compare it with screenshots/<page>.jpg.
4. Differences may only be data values; fix any layout, spacing, colour or font difference.
5. Mobile pages: compare at 390 px width with m_*.jpg.

4.2 Page → reference file

PAGE	ROUTE	REFERENCE HTML	SCREENSHOT
A Home (use v2)	/	preview/hero_photo.html	screenshots/hero_photo.jpg
B Explorer	/explore	preview/explorer.html	screenshots/explorer.jpg
C Calendar	/calendar/:region	preview/calendar.html	screenshots/calendar.jpg
D Sensor Lab	/sensors	preview/sensorlab.html	screenshots/sensorlab.jpg
E Data Studio	/studio	preview/studio.html	screenshots/studio.jpg
F Formula Lab	/formulas/:id	preview/formulas.html	screenshots/formulas.jpg
G Validation	/validation	preview/validation.html	screenshots/validation.jpg
H Source	/sources	preview/source.html	screenshots/source.jpg
I Blind Spot	/blind-spot	preview/blindspot.html	screenshots/blindspot.jpg
J Learn	/learn	preview/learn.html	screenshots/learn.jpg
K Workspace	/me	preview/workspace.html	screenshots/workspace.jpg
L Brief	/brief/:id	preview/brief.html	screenshots/brief.jpg
M Health	/sources/health	preview/health.html	screenshots/health.jpg

PAGE	ROUTE	REFERENCE HTML	SCREENSHOT
Mobile	all	preview/m_*.html	screenshots/m_*.jpg

4.3 Shared stylesheet (atlas.css) — copy exactly

```
:root {
  --space: #060A13;
  --ink: #0A1020;
  --panel: rgba(16, 24, 44, 0.74);
  --panel-solid: #111A2E;
  --line: rgba(150, 180, 255, 0.14);
  --line-strong: rgba(150, 180, 255, 0.28);
  --text: #EEF2FA;
  --dim: #9AA6BD;
  --faint: #67728A;
  --ember: #FF6A3D;
  --flame: #FFB547;
  --sky: #49C7F5;
  --modis: #8FB8FF;
  --viirs: #FF8A3D;
  --mint: #3DDC97;
  --rose: #FF4D6D;
  --violet: #9B8CFF;
  --paper: #F5F1E8;
  --paper-ink: #1B2233;
  --paper-line: #DCD3C2;
  --r-lg: 18px;
  --r-md: 12px;
  --r-sm: 8px;
}

* { box-sizing: border-box; margin: 0; padding: 0; }
html, body { background: var(--space); color: var(--text); font-family: "Inter", sans-serif; -we
    bkit-font-smoothing: antialiased; }
body.paper { background: var(--paper); color: var(--paper-ink); }
.mono { font-family: "JB", monospace; }
.caps { font-family: "SG", sans-serif; text-transform: uppercase; letter-spacing: 0.14em; font-s
    ize: 11px; font-weight: 700; }
.display { font-family: "Fraunces", serif; font-weight: 600; letter-spacing: -0.01em; }
.dim { color: var(--dim); }
.faint { color: var(--faint); }

.app { width: 1600px; height: 1000px; position: relative; overflow: hidden; background: radial-g
    radient(1200px 800px at 70% 40%, #0F1A33 0%, var(--space) 60%); }
.topbar { position: absolute; left: 0; right: 0; top: 0; height: 64px; display: flex; align-item
    s: center; padding: 0 24px; gap: 28px; border-bottom: 1px solid var(--line); background: rgb
    a(6, 10, 19, 0.6); backdrop-filter: blur(10px); z-index: 10; }
.brand { display: flex; align-items: center; gap: 12px; font-weight: 800; letter-spacing: 0.02em
    ; }
.brand small { display: block; font-family: "SG"; font-size: 10px; letter-spacing: 0.22em; color
    : var(--dim); font-weight: 500; }
.nav { display: flex; gap: 22px; font-size: 13.5px; color: var(--dim); }
.nav .on { color: var(--text); position: relative; }
.nav .on::after { content: ""; position: absolute; left: 0; right: 0; bottom: -22px; height: 2px
    ; background: var(--ember); border-radius: 2px; }
.spacer { flex: 1; }
.chip { display: inline-flex; align-items: center; gap: 6px; font-family: "SG"; font-size: 10.5p
    x; font-weight: 700; letter-spacing: 0.12em; padding: 5px 10px; border-radius: 999px; border
    : 1px solid var(--line-strong); color: var(--text); }
.chip .dot { width: 7px; height: 7px; border-radius: 50%; }
.chip.live .dot { background: var(--mint); box-shadow: 0 0 8px var(--mint); }
.chip.archive .dot { background: var(--sky); }
.chip.demo { border-color: rgba(255, 181, 71, 0.6); color: var(--flame); }
.chip.demo .dot { background: var(--flame); }
.chip.user .dot { background: var(--violet); }
.chip.cached .dot { background: var(--dim); }

.panel { background: var(--panel); border: 1px solid var(--line); border-radius: var(--r-lg); ba
```

```
    ckdrop-filter: blur(14px); box-shadow: 0 20px 50px rgba(0, 0, 0, 0.35); }
.panel-h { display: flex; align-items: center; justify-content: space-between; padding: 14px 16p
    x 10px; }
.panel-b { padding: 0 16px 16px; }
.btn { display: inline-flex; align-items: center; gap: 8px; font-weight: 600; font-size: 13px; p
    adding: 10px 16px; border-radius: 10px; border: 1px solid var(--line-strong); color: var(--t
    ext); background: rgba(255, 255, 255, 0.04); }
.btn.primary { background: var(--ember); border-color: var(--ember); color: #1A0A04; }
.btn.ghost { background: transparent; }
.seg { display: inline-flex; border: 1px solid var(--line-strong); border-radius: 10px; overflow
    : hidden; }
.seg span { font-family: "SG"; font-size: 11px; font-weight: 700; letter-spacing: 0.1em; padding
    : 7px 12px; color: var(--dim); }
.seg span.on { background: var(--text); color: var(--ink); }
.row { display: flex; align-items: center; gap: 10px; }
.kv { display: flex; justify-content: space-between; font-size: 12.5px; padding: 7px 0; border-b
    ottom: 1px dashed var(--line); }
.kv b { font-family: "JB"; font-weight: 700; }
.tile { padding: 14px 16px; border-radius: var(--r-md); border: 1px solid var(--line); backgroun
    d: rgba(255, 255, 255, 0.03); }
.tile .big { font-family: "Fraunces"; font-size: 34px; font-weight: 600; line-height: 1.1; }
.toggle { width: 34px; height: 19px; border-radius: 999px; background: var(--line-strong); posit
    ion: relative; flex: none; }
.toggle::after { content: ""; position: absolute; top: 3px; left: 3px; width: 13px; height: 13px
    ; border-radius: 50%; background: var(--dim); }
.toggle.on { background: rgba(255, 106, 61, 0.35); }
.toggle.on::after { left: 18px; background: var(--ember); }
.slider { height: 4px; border-radius: 4px; background: var(--line-strong); position: relative; m
    argin: 10px 0; }
.slider i { position: absolute; left: 0; top: 0; bottom: 0; background: var(--ember); border-rad
    ius: 4px; }
.slider b { position: absolute; top: -6px; width: 16px; height: 16px; border-radius: 50%; backgr
    ound: var(--text); border: 3px solid var(--ember); }
.legend-swatch { width: 10px; height: 10px; border-radius: 3px; display: inline-block; }
table.grid { border-collapse: collapse; width: 100%; font-size: 12.5px; }
table.grid th { text-align: left; font-family: "SG"; font-size: 10.5px; letter-spacing: 0.12em;
    text-transform: uppercase; color: var(--dim); padding: 8px 10px; border-bottom: 1px solid va
    r(--line-strong); }
table.grid td { padding: 9px 10px; border-bottom: 1px solid var(--line); }
.warn { border: 1px solid rgba(255, 181, 71, 0.45); background: rgba(255, 181, 71, 0.08); color:
     #FFD9A0; border-radius: 10px; padding: 10px 12px; font-size: 12.5px; }
.err { border: 1px solid rgba(255, 77, 109, 0.45); background: rgba(255, 77, 109, 0.08); color:
    #FFB3C1; border-radius: 10px; padding: 10px 12px; font-size: 12.5px; }
.ok { border: 1px solid rgba(61, 220, 151, 0.45); background: rgba(61, 220, 151, 0.08); color: #
    B5F5D8; border-radius: 10px; padding: 10px 12px; font-size: 12.5px; }

.mobile { width: 390px; min-height: 844px; position: relative; overflow: hidden; background: rad
    ial-gradient(500px 500px at 60% 30%, #0F1A33 0%, var(--space) 70%); }
.m-top { height: 56px; display: flex; align-items: center; padding: 0 16px; gap: 10px; border-bo
    ttom: 1px solid var(--line); }
.m-tabs { position: absolute; left: 0; right: 0; bottom: 0; height: 70px; display: flex; justify
    -content: space-around; align-items: center; border-top: 1px solid var(--line); background:
    rgba(6, 10, 19, 0.92); font-size: 10.5px; color: var(--dim); }
.m-tabs div { display: flex; flex-direction: column; align-items: center; gap: 4px; }
.m-tabs .on { color: var(--ember); }
.sheet { position: absolute; left: 0; right: 0; bottom: 70px; border-radius: 22px 22px 0 0; back
    ground: var(--panel-solid); border-top: 1px solid var(--line-strong); padding: 10px 16px 16p
    x; }
.sheet .grab { width: 40px; height: 4px; border-radius: 4px; background: var(--line-strong); mar
    gin: 0 auto 10px; }
```

⚠ Do not

Do not redesign, restyle or 'improve' the look. Do not swap fonts or colours. Do not replace the flat data globe with the realistic Earth (that is Home only). Do not use a UI kit that changes the look.

MEMORY

5 Keeping the AI's memory small and correct: graphify, data index, Antigravity

A big project does not fit in an AI's working memory at once. If Claude Code reads every data file and every document each session, it wastes its context and forgets details. Three tools fix this: a knowledge graph of the repository (graphify), one small index file that describes all data (DATA_INDEX.json), and short summary files the AI reads first. The rule for the AI is simple: read the index and the graph first, and open a raw file only when a task needs it.

5.1 graphify: a knowledge graph of the whole repository

graphify is an open-source skill for Claude Code. It reads code (with tree-sitter), Markdown docs, PDFs and images, and builds a graph of concepts and how they connect. It saves graph.json and a short GRAPH_REPORT.md, so later questions are answered from the graph without re-reading the sources. Re-runs only process changed files.

```
pip install graphifyy && graphify install     # package name has two y's; command is graphify
# macOS 'externally-managed-environment' error -> pipx install graphifyy

# inside Claude Code, at the repo root:
/graphify .                       # build the graph (code + docs/spec + docs/design screenshots)
/graphify . --update              # after changes: re-extract only changed files
/graphify query "which endpoint feeds the calendar page?"
/graphify path "resolver" "CalendarGrid"
graphify hook install             # rebuild after every git commit
```

Output goes to graphify-out/: graph.json (persistent graph), GRAPH_REPORT.md (main nodes and suggested questions), graph.html (interactive view), cache/. Add graphify-out/cache/ to .gitignore; keep GRAPH_REPORT.md committed so every session can read it.

5.2 DATA_INDEX.json: one file that describes all data

Raw data (millions of fire rows) must never be pasted into the AI. Instead data_kit/make_data_index.py (tested) scans data/ and writes DATA_INDEX.json with, for every CSV/Parquet file: path, size, columns and types, row count, date range and three sample rows. The AI reads this one small file to understand the data, then writes SQL (DuckDB) or code that runs on the real files. Run it after every pipeline run.

```
pip install duckdb
python data_kit/make_data_index.py      # writes DATA_INDEX.json
```

5.3 Files the AI reads first (in this order)

1. CLAUDE.md — rules and commands (short).
2. graphify-out/GRAPH_REPORT.md — map of the code and docs.
3. DATA_INDEX.json — what data exists, its columns and ranges.
4. data_kit/data_sources.json — every input x and its fallback sources.
5. docs/spec/CLAUDE_BUILD_SPEC.md — this spec; open the chapter for the current milestone only.
6. docs/design/preview/<page>.html + screenshots/<page>.jpg — only for the page being built.

5.4 If Claude Code is stuck or out of limit: Google Antigravity

Google Antigravity is an agent-first IDE (a VS Code fork) with a free preview tier for personal Gmail accounts (reported: Gemini Flash, about 50 completions a day, weekly limits; heavy users hit limits in a few hours — check current terms). It can also run Claude and Gemini Pro models and has built-in browser testing, which is useful for checking that pages match the screenshots. Use it as a backup, with the same files: open the repository, give it the same CLAUDE.md text as workspace rules, point it to docs/spec and docs/design, and ask it to continue the current milestone. graphify's README documents Claude Code; for other assistants, give them GRAPH_REPORT.md and DATA_INDEX.json as plain files.

⚠ Avoid

Do not let two agents edit the same files at the same time. Finish a milestone in one tool, commit, then switch. Always re-run tests after switching tools.

STACK

6 Repository, folders and technology

6.1 What already exists (FIREWORK repo)

Repository https://github.com/Dhruvsharma-DSAS/FIREWORK, live site https://firework-green.vercel.app/. Reuse: FIRMS live client (clients/firms.py), column and confidence normalisation, per-detection classifier (pipeline/classify.py), API with {data, meta.data_source}, React + Vite frontend with MapLibre globe, deck.gl and Recharts, 16 pytest files. Fix before judges: remove seed/evaluation_metrics.json (impossible F1 values), update README status, never show the synthetic seed corpus as results.

6.2 Target folder structure

```
FIREWORK/
  CLAUDE.md
  .env.example
  docs/spec/                this spec (PDF + MD)
  data_kit/                 data_sources.json, fx_engine.py, samples/
  pipeline/
    sources/                firms.py, gee.py, overpass.py, weather.py, orbits.py
    harmonize.py            (from SpaceApps_code.zip)
    accuracy_tools.py       (from SpaceApps_code.zip)
    calendar_build.py       (from SpaceApps_code.zip)
    failover.py, health_check.py
    run.py                  orchestrates the nightly job
  backend/app/
    main.py                 FastAPI app, CORS, routers
    db.py                   SQLModel engine (Supabase / Neon)
    models.py               tables (chapter 7)
    routers/                summary.py cells.py calendar.py calib.py validation.py studio.py br
    ief.py health.py ask.py
    services/               resolver.py (uses data_kit), ai_router.py, rag.py
  frontend/src/
    pages/                  Home Explorer Calendar SensorLab Studio FormulaLab Validation Sourc
    e BlindSpot Learn Workspace Brief Health
    components/             StatusChip LayerPanel CellInspector Timeline ValueWithBand FormulaC
    hip DemoBanner EmptyState
    hero/                   EarthHero (three.js, Home only)
    api/                    one file per endpoint
    styles/                 tokens.css
  tests/                    pytest + Playwright
  .github/workflows/        nightly.yml, test.yml
```

6.3 Approved technology (all free)

LAYER	CHOICE	BACKUP / NOTE
Frontend host	Cloudflare Pages	Vercel Hobby (non-commercial)

LAYER	CHOICE	BACKUP / NOTE
Frontend	React 19, Vite, React Router, MapLibre GL, deck.gl, Recharts, d3, satellite.js, three.js (hero only)	already in FIREWORK except three.js, satellite.js
Backend	FastAPI, Uvicorn, Pydantic, SQLModel/SQLAlchemy, GeoAlchemy2, httpx, tenacity	Hugging Face Spaces (Docker) or Render free (sleeps after 15 min)
Database	Supabase Postgres + PostGIS (500 MB free, pauses after 7 days idle)	Neon (1 GB/project) with same schema
Files	Parquet on Cloudflare R2 (10 GB, no egress fee), queried with DuckDB	GitHub Releases for small files
Pipeline	Python, pandas/polars, duckdb, earthengine-api, earthaccess, h3	runs on GitHub Actions cron or laptop
AI	router: Groq → OpenRouter free → Gemini Flash-Lite → GitHub Models → template	LiteLLM optional
RAG	pgvector in Supabase; local sentence-transformers or Gemini Embedding	Qdrant Cloud free (1 GB)
Monitoring	/api/health + uptime checker + Sentry free	-

DATA

7 Data sources, keys, pipeline and database

7.1 Environment variables (.env.example)

```
FIRMS_MAP_KEY=
EARTHDATA_USERNAME=
EARTHDATA_PASSWORD=
EE_PROJECT=
OPENAQ_API_KEY=
SUPABASE_URL=
SUPABASE_SERVICE_KEY=
DATABASE_URL=postgresql://...
NEON_DATABASE_URL=
R2_ACCOUNT_ID=
R2_ACCESS_KEY_ID=
R2_SECRET_ACCESS_KEY=
R2_BUCKET=firework-data
GROQ_API_KEY=
OPENROUTER_API_KEY=
GEMINI_API_KEY=
GITHUB_MODELS_TOKEN=
```

7.2 Where to get the keys

KEY	SIGN-UP	FREE LIMIT
FIRMS_MAP_KEY	https://firms.modaps.eosdis.nasa.gov/api/map_key/	5,000 transactions / 10 min; 1–5 days per call
Earthdata	https://urs.earthdata.nasa.gov/users/new	bulk downloads
EE_PROJECT	https://earthengine.google.com/noncommercial/	Community tier 150 EECU-hours/month (since 27 Apr 2026)
OPENAQ_API_KEY	https://explore.openaq.org/register	60/min, 2,000/hour
Supabase	https://supabase.com	500 MB DB, 1 GB files, 5 GB egress
Groq	https://console.groq.com/keys	e.g. 1,000 requests/day on 70B models
OpenRouter	https://openrouter.ai	free models: 20/min, 50/day
Gemini	https://aistudio.google.com/apikey	Flash/Flash-Lite only; limits shown in AI Studio

7.3 Formula inputs (x) and their sources, in fallback order

INPUT	MEANING	FORMULAS	SOURCES (PRIMARY → BACKUPS)
x_fire_points	Active-fire detections (MODIS + VIIRS)	F01, F02, F03, F05, F06, F07, F08, F13, F20, F21, F44	FIRMS Area API → FIRMS archive download (Download Request) → LAADS DAAC / earthaccess (MCD14ML, VNP14IMG, VJ114IMG) → GEE community catalog FIRMS vectors
x_clear_days	Days each cell was visible (not cloud) - daily FireMask	F09, F25	Google Earth Engine MOD14A1 / MYD14A1 / VNP14A1 → Microsoft Planetary Computer modis-14A1-061 → NASA AppEEARS area request
x_burned_area	Burned area per month (blind-spot check)	F46	GEE MODIS/061/MCD64A1 → ORNL DAAC MODIS subsets (MCD64A1) → ESA FireCCI
x_land_cover	Cropland / tree / built-up class around a fire	F08, F42, F43	ESA WorldCover 10 m (GEE) → ESA WorldCover on AWS S3 → MODIS MCD12Q1 yearly land cover
x_industry	Industrial sites, kilns, flares near a fire	F08, F42, F43	OSM Overpass API → Overpass mirror (private.coffee / maps.mail.ru) → Geofabrik India extract (.osm.pbf) → VIIRS Nightfire / EOG gas flares
x_weather	Wind, humidity, rain, temperature for context	context panels	Open-Meteo Forecast + Archive → NASA POWER daily point → Copernicus CDS ERA5
x_air_quality	PM2.5 near the burning (impact panel)	impact panel	OpenAQ API v3 → Open-Meteo Air Quality (CAMS model) → CPCB / WAQI
x_orbits	Satellite orbit elements → overpass times	F01, F45	CelesTrak GP API → Space-Track → Derive from FIRMS acq_time
x_boundaries	District / state boundaries for regions and briefs	F04 (areas), region totals	geoBoundaries API → Natural Earth → GADM
x_ndvi	Vegetation greenness / harvest timing	season context (F29)	GEE MODIS/061/MOD13Q1 → ORNL MODIS subsets → Sentinel-2 via Earth Search STAC
x_granule_index	Which product files exist for a date (health + archive)	pipeline / health page	NASA CMR Search API → earthaccess Python library

The registry file data_kit/data_sources.json holds the same list with keys, limits and URLs. backend/app/services/resolver.py must use data_kit/fx_engine.py logic: try sources in order, skip a source that failed in the last 10 minutes, run quality gates, cache, and return {data, meta} with data_source, fallback_used and tried.

7.4 What the main APIs return

API	CALL	FIELDS WE USE
FIRMS VIIRS CSV	/api/area/csv/{KEY}/VIIRS_NOAA20_NRT/W,S,E,N/{1-5}	latitude, longitude, bright_ti4, bright_ti5, scan, track, acq_date, acq_time, satellite, confidence (l/n/h), version, frp, daynight
FIRMS MODIS CSV	/api/area/csv/{KEY}/MODIS_NRT/W,S,E,N/{1-5}	brightness, bright_t31, confidence (0–100), satellite Terra/Aqua; archive adds type (0 veg, 1 volcano, 2 static, 3 offshore)
GEE MYD14A1/ MOD14A1/VNP14A1	FireMask band, monthly reduceRegions over cells	FireMask 4 cloud, 5 clear land, 7–9 fire → clear_days, fire_days
GEE MCD64A1	BurnDate band, monthly	BurnDate > 0 → burned pixels × 0.25 km^2
GEE WorldCover v200	frequencyHistogram per cell	40 cropland, 50 built-up, 10 trees

API	CALL	FIELDS WE USE
Overpass	landuse=industrial, man_made=kiln in bbox	elements[].center lat/lon, tags
Open-Meteo archive	daily wind, humidity, rain	daily.time[], daily.<var>[]
OpenAQ v3	/v3/locations?coordinates=lat,lon&radius=25000	sensors[].parameter pm25, measurements value
CelesTrak GP JSON	gp.php?CATNR=43013&FORMAT=json	EPOCH, MEAN_MOTION, INCLINATION … (NORAD 37849 S-NPP, 43013 N20, 54234 N21, 25994 Terra, 27424 Aqua)

Full field-by-field responses are in FIREWORK_APIs_Keys_Datasets.pdf and data_kit/samples/. Sample values there are illustrative; data_kit/fetch_samples.py saves real responses with your keys.

7.5 Pipeline (pipeline/run.py)

1. Ingest FIRMS archive CSVs (data/raw) + NRT via API → normalise columns (harmonize.normalise_firms).
2. Filter: equal confidence (MODIS ≥ 30, VIIRS n/h), scan ≤ 2 km (F02, F07).
3. Mask static sources: persistence over 3 years > 0.20 (F08) + flare list.
4. Grid to 0.1° cells and cell-days (F03, F05) → Parquet data/cell_day/sensor=*/year=*.
5. Earth Engine: monthly clear_days and fire_days per cell per sensor (F09) → Parquet.
6. Monthly counts and rates (F06, F09); overlap ratios 2012–2025 (F10).
7. Calibration k per region × month × day/night with CI: median (F11), Poisson GLM (F14), EB shrinkage (F18); store in table calib.
8. Harmonise MODIS years (F12) + bands (F24 bootstrap, F25 Monte Carlo).
9. Calendar: baselines, robust z, season dates, MK + Sen trend, BH correction (F27–F32, F37).
10. Validation: LOYO, Pettitt, PELT, seam ratio, Bland–Altman (F33–F36, F23) → table validation.
11. Flags: burned-area blind-spot ratio (F46), diurnal fraction (F45), source context (F42–F44).
12. Upload Parquet to R2; upsert small tables to Supabase; write lineage row.

7.6 Database schema (Supabase / Neon, PostGIS)

```
create extension if not exists postgis;
create extension if not exists vector;

create table cell (
  cell text primary key,          -- '1209_2558' (0.1 deg)
  geom geometry(Polygon, 4326),
  area_km2 real,
  district text, state text
);
create table cell_month (
  cell text references cell(cell),
  month date,
  sensor text,                    -- MODIS_T, MODIS_A, VIIRS_SNPP, VIIRS_N20, VIIRS_N21
  dn char(1),                     -- D or N
  fire_days int, clear_days int, rate real,
  primary key (cell, month, sensor, dn)
);
create table calendar (
  region text, month date,
  h real, lo real, hi real,       -- harmonised VIIRS-eq cell-days + 90% band
  raw_modis real, raw_viirs real,
  z real, flags text[],
```

```
  version text,
  primary key (region, month, version)
);
create table calib (
  region text, month int, dn char(1),
  k real, k_lo real, k_hi real, n int, method text, aic real, version text,
  primary key (region, month, dn, version)
);
create table validation (region text, test text, value real, passed boolean, version text, detai
    l jsonb);
create table lineage (run_id text primary key, started timestamptz, inputs jsonb, code_sha text,
     params jsonb, sources_used jsonb);
create table user_dataset (id uuid primary key, owner uuid, name text, rows int, mapping jsonb,
    created timestamptz);
create table doc_chunk (id bigserial primary key, source text, heading text, body text, embeddin
    g vector(384));
```

FORMULAS

8 All 46 formulas to implement

Each formula has a full page (purpose, derivation, Hinglish explanation, worked example) in the A4 atlas. Here is what Claude Code needs: the equation, the inputs and where it lives. Status: E = existing code in SpaceApps_code.zip, T = tested there, P = to write.

8.1 Time, space and counting

NO	FORMULA	EQUATION	CODE HOOK	ST
F01	Local solar time of an overpass	$t_{\text{LST}} = t_{\text{UTC}} + \frac{\lambda}{15^\circ\,\text{h}^{-1}}$	lst = (utc_h + lon/15) % 24	E
F02	Pixel growth with scan angle	$\frac{s(\theta)}{s_0} \approx \frac{1}{\cos^2\theta}$ (along scan), $\frac{t(\theta)}{t_0} \approx \frac{1}{\cos\theta}$ (along track)	keep = df.scan <= 2.0	P
F03	Grid cell index	$i = \left\lfloor \frac{\varphi + 90}{\Delta} \right\rfloor, \quad j = \left\lfloor \frac{\lambda + 180}{\Delta} \right\rfloor$	harmonize.cell_id(lat, lon, 0.1)	E
F04	Area of a grid cell on the sphere	$A = R^2\,\Delta\lambda\,(\sin\varphi_2 - \sin\varphi_1)$	R*R*dlon*(sin(p2)-sin(p1))	P
F05	Fire cell-day indicator	$c_{g,d} = \mathbb{1}\left[n_{g,d} \geq 1\right]$	harmonize.to_cell_days(df, 0.1)	E
F06	Monthly cell-day count	$C_{g,m} = \sum_{d \in m} c_{g,d}$	harmonize.monthly_counts	E
F07	Equal-confidence filter	$\text{keep}_i = \mathbb{1}[q_i \geq \tau_s], \quad \tau_{\text{MODIS}} = 30, \quad \tau_{\text{VIIRS}} = \text{nominal}$	normalise_firms(df, sensor)	P
F08	Persistence (static-source mask)	$p_g = \frac{\text{active days}_g}{\text{total days}}$	harmonize.persistence(a, t)	E
F09	Clear-sky fire rate	$r_{g,m} = \frac{F_{g,m}}{K_{g,m}}$	clear_sky_rate(F, K)	T

8.2 Calibration between sensors

NO	FORMULA	EQUATION	CODE HOOK	ST
F10	Monthly overlap ratio	$k_m = \frac{V_m}{M_m} \quad (m \in \text{overlap 2012–2025})$	harmonize.overlap_ratio	E
F11	Median of ratios	$\hat{k} = \text{median}\{k_m\}$	statistics.median(ratios)	ET
F12	Harmonised MODIS value	$H_{g,m} = \hat{k} \cdot M_{g,m} \quad \text{for } m < 2012$	harmonise_modis(m, k)	E
F13	Separate day and night factors	$\hat{k}_{\text{day}} = \text{median}\,\frac{V_m^D}{M_m^D}, \quad \hat{k}_{\text{night}} = \text{median}\,\frac{V_m^N}{M_m^N}$	df.groupby('daynight')	P
F14	Poisson GLM with exposure offset	$\log \mathbb{E}[V_m] = \log M_m + \beta, \quad \hat{k} = e^{\hat{\beta}}$	sm.GLM(V, ones, family=Poisson(), offset=log(M))	T
F15	Negative-binomial variance	$\text{Var}(V) = \mu + \frac{\mu^2}{\theta}$	sm.families.NegativeBinomial	P

NO	FORMULA	EQUATION	CODE HOOK	ST
F16	Delta-method CI for a ratio	$\mathrm{Var}\left(\log \frac{V}{M}\right) \approx \frac{1}{V} + \frac{1}{M}$	se = sqrt(1/V + 1/M)	P
F17	Shrink local factor toward global	$\tilde{k} = \frac{n\,k_{\mathrm{loc}} + n_0\,k_{\mathrm{glob}}}{n + n_0}$	shrink_ratio(k, n, kg, n0)	E
F18	Empirical Bayes shrinkage of log k	$\widehat{\log k_r} = w_r\,\overline{\log k_r} + (1 - w_r)\,\mu, \quad w_r = \frac{\tau^2}{\tau^2 + s_r^2/n_r}$	empirical_bayes_log_ratio	T
F19	Power law vs ratio (AIC)	$V = a\,M^b$ vs $V = kM, \quad \mathrm{AIC} = n \ln \frac{\mathrm{RSS}}{n} + 2p$	fit_power_law, aic_gaussian	T
F20	Detection-probability curve (FRP50)	$P(\text{MODIS detects} \mid \mathrm{FRP}) = \frac{1}{1 + e^{-(\mathrm{FRP} - \mathrm{FRP}_{50})/s}}$	detection_curve	T
F21	Log transform of FRP	$y = \ln(1 + \mathrm{FRP})$	np.log1p(frp)	P
F22	Quantile mapping	$x' = F_V^{-1}(F_M(x))$	quantile_map	T
F23	Bland–Altman agreement	$\bar{d} = \overline{a - b}, \quad \mathrm{LoA} = \bar{d} \pm 1.96\,\mathrm{SD}(a - b)$	bland_altman(a, b)	T

8.3 Uncertainty

NO	FORMULA	EQUATION	CODE HOOK	ST
F24	Block bootstrap over years	$\hat{\theta}^{*(b)} = T(\text{years resampled with replacement}), \ \mathrm{CI} = [q_{2.5}, q_{97.5}]$	block_bootstrap_years	T
F25	Monte Carlo error propagation	$H^{(s)} = k^{(s)} M^{(s)}, \quad k^{(s)} \sim \mathcal{N}(\hat{k}, \sigma_k^2), \ M^{(s)} \sim \mathrm{Poisson}(M)$	numpy.random.default_rng(0)	T
F26	Exact Poisson CI (Garwood)	$\left[\frac{1}{2}\chi^2_{2n,\alpha/2}, \ \frac{1}{2}\chi^2_{2(n+1),1-\alpha/2}\right]$	scipy.stats.chi2.ppf	P

8.4 Anomalies and seasons

NO	FORMULA	EQUATION	CODE HOOK	ST
F27	Robust z-score	$z = \frac{\ln(1+x) - \mathrm{med}}{\max(1.4826\,\mathrm{MAD},\ \sigma_{\min})}$	harmonize.robust_z	ET
F28	Month baseline and spread	$\mathrm{base}_m = \mathrm{median}_y\, H_{y,m}, \quad \mathrm{spread}_m = 1.4826\ \mathrm{MAD}_y\, H_{y,m}$	calendar_build.month_ baseline	E
F29	Season start and end (10% / 90%)	$t_{\mathrm{start}} = \min\{t : S(t) \geq 0.10\,S_{\mathrm{tot}}\}, \quad t_{\mathrm{end}} = \min\{t : S(t) \geq 0.90\,S_{\mathrm{tot}}\}$	season_dates	E

8.5 Trends and seam tests

NO	FORMULA	EQUATION	CODE HOOK	ST
F30	Mann–Kendall trend statistic	$S = \sum_{i<j} \mathrm{sgn}(x_j - x_i)$	pymannkendall.hamed_rao_ modification_test	T

NO	FORMULA	EQUATION	CODE HOOK	ST
F31	Hamed–Rao autocorrelation correction	$\mathrm{Var}^*(S) = \mathrm{Var}(S)\cdot\frac{n}{n_s^*},\quad \frac{n}{n_s^*} = 1+\frac{2}{n(n-1)(n-2)}\sum_k (n-k)(n-k-1)(n-k-2)\rho_k$	pymannkendall	T
F32	Sen's slope	$\beta = \mathrm{median}_{i<j}\,\frac{x_j - x_i}{j-i}$	harmonize.sen_slope	E
F33	Pettitt change-point test	$U_t = \sum_{i\le t}\sum_{j>t}\mathrm{sgn}(x_j - x_i),\quad K = \max_t \lvert U_t\rvert,\quad p \approx 2e^{-6K^2/(n^3+n^2)}$	accuracy_tools.pettitt	T
F34	PELT multiple change points	$\min_{\tau}\sum_i \mathcal{C}\left(x_{(\tau_{i-1}+1):\tau_i}\right) + \beta\lvert\tau\rvert$	rpt.Pelt(model='rbf').fit(x).predict(pen)	P
F35	Seam ratio	$R_{\text{seam}} = \frac{\lvert H_{\text{after}} - H_{\text{before}}\rvert}{\lvert V_{\text{after,raw}} - M_{\text{before}}\rvert}$	seam_ratio	E
F36	Leave-one-year-out (LOYO) error	$\mathrm{MAPE}_{\text{LOYO}} = \frac{1}{Y}\sum_y \frac{\lvert \hat{k}_{-y} M_y - V_y\rvert}{V_y}$	for y in years: fit(exclude=y)	P
F37	Benjamini–Hochberg FDR	reject $H_{(i)}$ for $i \le \max\left\{i : p_{(i)} \le \frac{i}{m}\alpha\right\}$	benjamini_hochberg	T

8.6 Fusion and AI

NO	FORMULA	EQUATION	CODE HOOK	ST
F38	Kalman fusion of two sensors	$\hat{x} = \hat{x}^- + K(z - \hat{x}^-),\qquad K = \frac{P^-}{P^- + R}$	kalman_fuse	T
F39	Split conformal prediction	$\hat{q} = \mathrm{Quantile}_{\lceil (n+1)(1-\alpha)\rceil/n}\{1-\hat{p}_{y_i}(x_i)\},\quad C(x) = \{y : 1-\hat{p}_y(x) \le \hat{q}\}$	split_conformal_sets	T
F40	Haversine distance	$d = 2R\arcsin\sqrt{\sin^2\frac{\Delta\varphi}{2} + \cos\varphi_1\cos\varphi_2\sin^2\frac{\Delta\lambda}{2}}$	sklearn haversine metric	P
F41	DBSCAN fire clustering	$\mathrm{core}(p) \iff \lvert N_\varepsilon(p)\rvert \ge \mathrm{minPts},\quad N_\varepsilon(p) = \{q : d(p,q) \le \varepsilon\}$	DBSCAN(eps=..., metric='haversine')	P

8.7 Source attribution and blind spot

NO	FORMULA	EQUATION	CODE HOOK	ST
F42	Distance-decay exposure (IE / AE)	$\mathrm{IE}_g = \sum_{f\in\text{industry}} e^{-d_{g,f}/L},\qquad \mathrm{AE}_g = \sum_{c\in\text{crop}} e^{-d_{g,c}/L}$	np.exp(-d/L).sum()	P
F43	Context log-ratio L	$L_g = \ln\frac{\mathrm{IE}_g + 1}{\mathrm{AE}_g + 1}$	harmonize.l_context	E
F44	Night fraction	$\nu_g = \frac{N_g^{\text{night}}}{N_g^{\text{day}} + N_g^{\text{night}}}$	(dn=='N').mean()	P
F45	Diurnal fraction seen by overpasses	$f_{\text{seen}} = \sum_o \int_{t_o-w}^{t_o+w} \frac{1}{\sigma\sqrt{2\pi}} e^{-(t-\mu)^2/2\sigma^2}\,dt$	diurnal_fraction_seen	T
F46	Blind-spot ratio B	$B_{g,y} = \frac{\mathrm{BA}_{g,y}/C_{g,y}}{\mathrm{median}_{y'<2021}\left(\mathrm{BA}_{g,y'}/C_{g,y'}\right)}$	ba/c / median(ba/c pre2021)	P

⚠ Must-pass tests

clear_sky_rate(30,600)=0.05 and (5,0)=None; EB gives the 2-month region the lowest weight; robust z beats classic z on an outlier history; Pettitt finds the synthetic break at year 10 and not after harmonisation; BH reduces 94 to 53 on the synthetic set; diurnal fraction 31.6% (peak 13:00) vs 3.3% (peak 16:18). These

are synthetic checks, not results.

API

9 Backend API contract

Every endpoint returns JSON {data, meta}. meta always contains data_source (string or object per input), version, generated_at (ISO UTC), units, demo (bool) and fallback_used (bool). Errors return {data: null, meta: {..., error}} with a proper HTTP status.

METHOD + PATH	RETURNS	PAGE
GET /api/summary/global	hex counts last 7 days	Home
GET /api/cells?bbox=W,S,E,N&date=YYYY-MM&res=5	aggregated cells for map	Explorer
GET /api/cells/{cell}?month=YYYY-MM	inspector: fire_days, clear_days, rate, h, lo, hi, source mix, flags, overpasses	Explorer
GET /api/calendar/{region}?mode=raw|harmonised	matrix years × months with lo/hi, z, flags	Calendar
GET /api/calib/{region}	k, CI, method, AIC, day/night	Sensor Lab
GET /api/validation/{region}	LOYO, seam, BA, coverage results + pass rules	Validation
GET /api/formulas/{id}	formula metadata + live example inputs	Formula Lab
GET /api/source/{cell}	persistence, night fraction, L, land cover, class + conformal set	Source
GET /api/blindspot/{region}	diurnal curve, overpass windows, B ratio	Blind Spot
POST /api/studio/upload	schema guess + preview rows	Studio
POST /api/studio/compare	match table + map + lineage	Studio
GET /api/brief/{region}?season=YYYY	numbers JSON + plain-language text	Brief
POST /api/ask	RAG + tools answer with citations	Ask helper
GET /api/health	status per data source + last update	Health


```
{
  "data": {"cell": "1209_2558", "month": "2024-11", "fire_days_viirs_eq": 14, "lo": 12, "hi": 16
    ,
          "clear_days": 22, "rate": 0.64, "unit": "VIIRS-equivalent fire cell-days", "flags": [
    "blind_spot"]},
  "meta": {"data_source": {"fire_points": "FIRMS archive VNP14IMG + MCD14ML", "clear_days": "GEE
     MYD14A1",
          "burned_area": "GEE MCD64A1"}, "fallback_used": false, "version": "calib-0.3",
          "generated_at": "2026-10-20T06:00:00Z", "units": "VIIRS-eq cell-days", "demo": true}
}
```

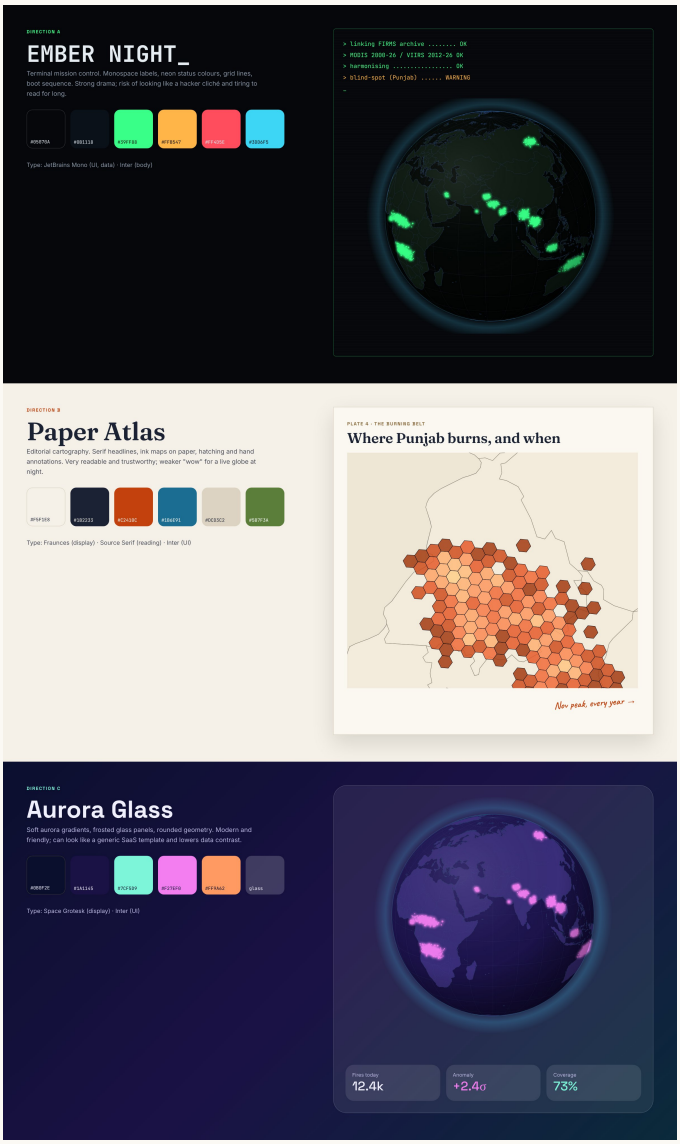
WEBSITE

10 Design system and the 13 pages

10.1 Design tokens (frontend/src/styles/tokens.css)

```
:root {
  --space: #060A13; --panel: rgba(16,24,44,0.74); --line: rgba(150,180,255,0.14);
  --text: #EEF2FA; --dim: #9AA6BD;
  --ember: #FF6A3D; --flame: #FFB547; --sky: #49C7F5;
  --modis: #8FB8FF; --viirs: #FF8A3D; --mint: #3DDC97; --rose: #FF4D6D; --violet: #9B8CFF;
  --paper: #F5F1E8;
  --r-lg: 18px; --r-md: 12px; --r-sm: 8px;
}
/* fonts: Fraunces (display), Inter (UI), JetBrains Mono (numbers), Space Grotesk (labels), Cave
   at (hand notes) */
```

- Status chips: LIVE (mint), ARCHIVE (sky), CACHED (grey), DEMO (flame), USER (violet) — driven by meta.
- MODIS is always blue, VIIRS always orange; never colour-only (add labels/shapes).
- Motion: 120 ms hover, 240 ms panels, 600 ms timeline step, 1,200 ms camera flights; respect prefers-reduced-motion.
- Breakpoints: ≥1440 three columns; 1024–1439 panel collapses; 768–1023 inspector drawer; <768 bottom sheet + 5-tab bar.
- Realistic three.js Earth ONLY on the Home hero. All data pages use the flat MapLibre + deck.gl globe.



Design directions studied; chosen system 'Mission Atlas' = dark mission-control canvas + editorial type.

10.2 Page A · Cinematic Homepage (/)

Objective: Give a first-time visitor the purpose, the honesty promise and three clear ways in, within 10 seconds. Audience: Everyone: judges, students, journalists, officials, researchers.

The Home page is the front door of FIREWORK CHRONOS. A first-time visitor — a judge, a student, a pollution-board officer — should understand in about ten seconds what the site does, why it matters now, and where to click next. The headline states the promise (25 years of fire on one honest ruler), the kicker names the NASA challenge, and three buttons lead to the three main jobs: explore the Earth, open the burning calendar, or upload your own data.

Two small chips in the top bar, LIVE and ARCHIVE, are always visible. They are read from the API's meta field, so if the live FIRMS feed is down the chip turns grey and says CACHED instead of pretending. The 'Why now' panel gives the three mission end dates with sources, because the urgency of the project comes from those dates: once MODIS stops, the morning view and the older half of the record can no longer be checked against a second instrument.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Version 2 (use this for Home): realistic night Earth (NASA Blue/Black Marble textures), sunrise flare at the limb, satellite moving on a 24 s orbit loop, original top bar with LIVE/ARCHIVE chips kept. Code: Home_hero_kit.zip (earth3d.js). Path: quadratic Bézier in camera space P0(−0.15,1.75) → P1(0.70,1.50) → P2(2.20,0.95), depth −4.8; pause when tab hidden; poster image for LCP and reduced motion.



Home v2 hero (three.js render).

Marked elements

1. Brand: chrono-orbit mark + name
2. Primary navigation (10 sections)
3. Data status chips: LIVE, ARCHIVE (always visible)
4. Challenge kicker
5. Three entry points: explore, calendar, upload
6. Record facts: length, sensors, the 3 tags every value carries
7. Why now: mission end dates with sources
8. Honesty label: demo pattern, Natural Earth

Components

- Hero globe (MapLibre globe or WebGL hero layer)
- Headline in Fraunces, tagline in Inter
- CTA group
- Stat tiles
- Why-now panel
- Status chips

Data

- Last NRT fetch time (FIRMS)
- Archive coverage years

- Mission dates (static, cited)
- Lightweight global hex summary (pre-computed JSON, ~200 kB)

Interactions

- Globe auto-rotates 3.5°/s; pauses on hover or drag; resumes after 10 s
- CTA hover: 120 ms lift
- Reduced motion: static globe image

States

- Loading: static globe poster + skeleton tiles
- Live feed down: LIVE chip turns grey CACHED with timestamp
- Hex summary missing: globe without fires + note

Frontend notes

- Route '/'; reuse existing MapLibre v6 globe in MapCanvas (already supports projection globe + rotation)
- Lazy-load the map after first paint; poster image for LCP

Backend

- GET /api/summary/global → hex counts last 7 days + meta.data_source
- GET /api/health → chip states

Limits

- Fire points on the hero are a summary, not every detection
- Never animate demo data as if live

Done when (acceptance)

- LCP < 2.5 s on 4G laptop profile
- Status chips always match /api/health
- Keyboard can reach all 3 CTAs in order

10.3 Page B · Earth Explorer (Globe Workspace) (/explore)

Objective: Let users find any place, see where and when burning was observed, and inspect one cell with its evidence. Audience: Researchers, officials, journalists, students.

The Explorer is the main working screen. It shows the globe with fire detections and lets the user move from the whole planet down to one 0.1° cell (about 106 km^2 in Punjab). On the left, the layer panel decides what is drawn: VIIRS and MODIS hot-spots, H3 density hexagons, burned area, land cover, clear-sky coverage, the day/night line and satellite ground tracks. One confidence rule is applied to both sensors, so switching sensors never changes the strictness of the filter.

Clicking a cell opens the inspector on the right. It shows the fire cell-days for the chosen month, how many days the cell was actually visible (clear-sky), the rate, the harmonised value with its 90% band, the source mix and, if needed, a blind-spot warning. Below it, a small card lists which satellites pass over the cell and at what local time, next to the hour when burning usually peaks. The timeline at the bottom plays day by day, month by month or year by year and marks the 2012 seam where VIIRS begins.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. Layer panel: fire, context, colour mode, same confidence rule for both sensors
2. Place search with bbox (W,S,E,N)
3. Cell inspector: counts, clear-sky, rate, harmonised value with band, source mix, blind-spot warning
4. Sensors over this cell: overpass times vs burning peak
5. Timeline: play by day/month/year; MODIS era vs VIIRS era; 2012 seam marked

Components

- Globe/map canvas (MapLibre v6 + deck.gl overlay)
- Layer panel
- Search
- Inspector
- Overpass card
- Timeline scrubber

Data

- Hex/cell aggregates per day or month (vector tiles)
- Raw detections only when zoomed in ($\leq$ 5,000 features per request)
- Clear-sky counts
- Burned area raster tiles
- TLE for ground tracks (CelesTrak)

Interactions

- Zoom thresholds: z<4 H3 res 2–3 hexes; 4–7 res 5; >7 raw points
- Click cell → camera eases 1,200 ms; inspector slides 240 ms
- Timeline play: 1 step / 600 ms; cross-fade 250 ms

States

- Loading tiles: shimmer on hex layer, inspector skeleton
- Empty: 'No detections in this window' with nearest active month
- Unavailable layer: toggle disabled + reason
- Too many points: auto-switch to hex mode

Frontend notes

- Extend existing MapCanvas.jsx and LayerPanel.jsx
- deck.gl H3HexagonLayer for density; ScatterplotLayer (already used) for points
- URL state: /explore?bbox=...&date=...&layers=...

Backend

- GET /api/cells?bbox&date&res → aggregated cells
- GET /api/cells/{id} → inspector payload with meta
- Vector tiles via ST_AsMVT (FIREWORK rule)

Limits

- A hotspot is a pixel centre, not a fire outline
- Rate needs clear-sky data; shown as '—' if missing

Done when (acceptance)

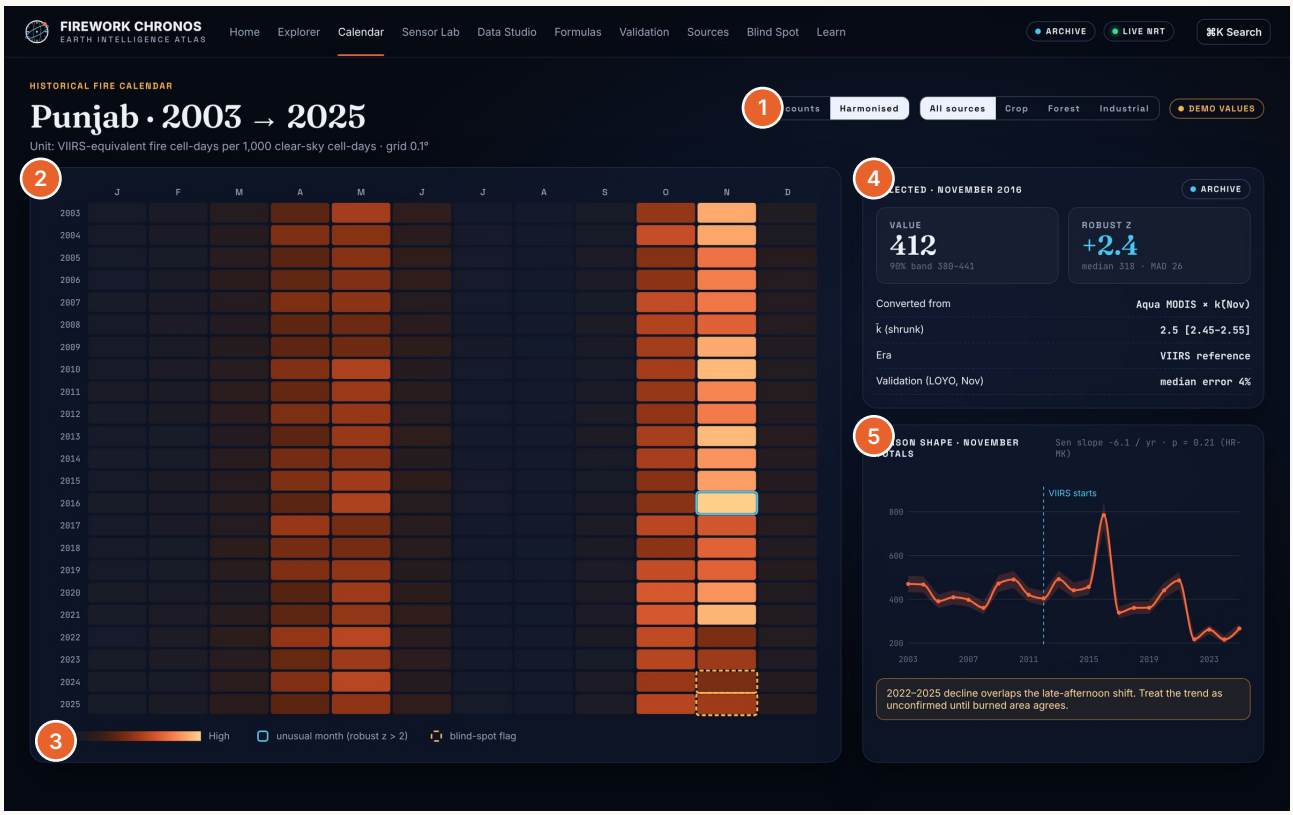
- 60 fps pan at 100k aggregated cells on a mid laptop
- Inspector numbers equal the calendar value for the same cell-month
- All layers list source + version on hover

10.4 Page C · Historical Fire Calendar (/calendar/:region)

Objective: Show 25 years × 12 months of burning on one consistent scale, with unusual months, seasons and uncertainty. Audience: Officials, scientists, journalists.

The Historical Fire Calendar is the answer to the core of the challenge: a table of years by months that shows how much burning happened in a region, on one consistent scale from 2000 to today. Every cell is a colour; darker means more VIIRS-equivalent fire cell-days. The user can switch between Raw (what each sensor counted) and Harmonised (MODIS converted with the factor k), and the visible jump at 2012 in Raw mode disappears in Harmonised mode — that is the 'aha' moment of the demo.

Clicking a month shows its value with an uncertainty band, how unusual it is compared with the same month in other years (robust z-score), how it was converted, and how well the method predicted that kind of year in leave-one-year-out tests. A small chart under the calendar shows the season shape for November, the start of the VIIRS era and the long-term trend with its significance. Values built from demo data carry a DEMO label until real archive numbers replace them.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. Toggles: Raw ↔ Harmonised; source split; DEMO label on mock values
2. Calendar heatmap (years × months)
3. Legend: unusual month (robust z > 2) and blind-spot flag
4. Selected cell: value, band, z, how it was converted, LOYO error
5. Season shape: November totals with band, VIIRS start line, trend and warning

Components

- Heatmap (D3 or custom SVG)
- Mode toggles
- Detail panel
- Mini line chart (Recharts, already in repo)

Data

- Harmonised matrix H[y,m] with lo/hi
- Raw matrix
- z-scores
- Season dates
- Trend test results
- Flags

Interactions

- Raw→Harmonised: 600 ms colour tween so the 2012 step visibly disappears
- Hover: tooltip with value and band
- Click: detail panel

States

- Region with few fires: grey cells + 'indicative only'
- Archive not processed: calendar locked with ETA
- Loading: grey grid skeleton

Frontend notes

- New page Calendar.jsx (CalendarGrid component drafted in Book 4)
- Colour scale single hue; colour-blind safe

Backend

- GET /api/calendar/{region} with mode = raw or harmonised, source = all, crop, ...
- Response carries unit, years, validation, flags

Limits

- Counts ≠ burned area ≠ emissions
- A trend with $p > 0.05$ is 'not shown', not 'no change'

Done when (acceptance)

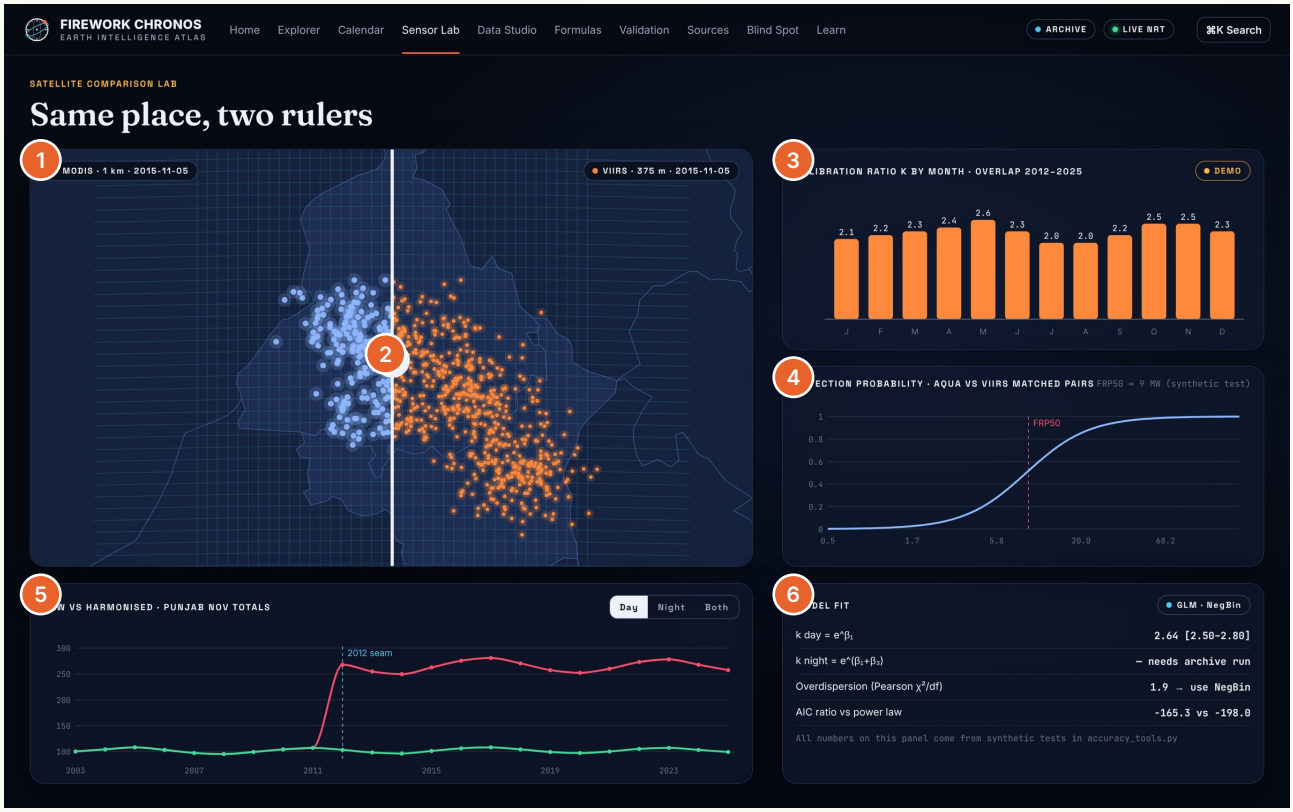
- Toggle changes only colours, not layout
- Every cell tooltip shows unit and band
- Same numbers as API and CSV export

10.5 Page D · Satellite Comparison Lab (/sensors)

Objective: Make the MODIS vs VIIRS difference visible and show how the calibration ratio is learned. Audience: Researchers, technical reviewers, students.

The Sensor Lab explains why MODIS and VIIRS disagree and how we correct it. It is the page a scientist or a curious student opens to see the evidence behind the conversion factor. The same fires are shown through both sensors side by side, with a swipe handle, so the user sees with their own eyes that VIIRS draws many small dots where MODIS draws a few large ones.

Under the map, a scatter plot compares monthly VIIRS and MODIS cell-days in the overlap period, with the fitted ratio and, if chosen, the power-law curve. Controls let the user split day and night, change the confidence threshold, or see the detection curve that shows at what fire power MODIS starts to miss fires. Every chart links to its formula page, so nothing is a black box.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. Swipe map: MODIS left, VIIRS right, same day
2. Swipe handle
3. Calibration ratio k by month (overlap 2012–2025)
4. Detection probability curve and FRP50
5. Raw vs harmonised series with day/night switch
6. Model fit panel: k day, k night, overdispersion, AIC

Components

- Swipe compare (two synced maps)
- Bar chart
- Logistic curve
- Line chart
- Model panel

Data

- Same-day detections by sensor
- k by region-month with CI
- Matched-pair FRP table
- GLM outputs

Interactions

- Swipe handle drag; keyboard arrows move 5%
- Day/Night/Both switch re-draws in 300 ms

States

- Night k not computed: '— needs archive run'
- No matched pairs: curve hidden + explanation

Frontend notes

- Two MapLibre instances with synced camera, or one map + clip mask
- Recharts for bars/lines

Backend

- GET /api/calibration/{region}
- GET /api/detection-curve/{region}

Limits

- Matched pairs need near-simultaneous overpasses ($\leq$ 15 min)
- Terra drift years excluded

Done when (acceptance)

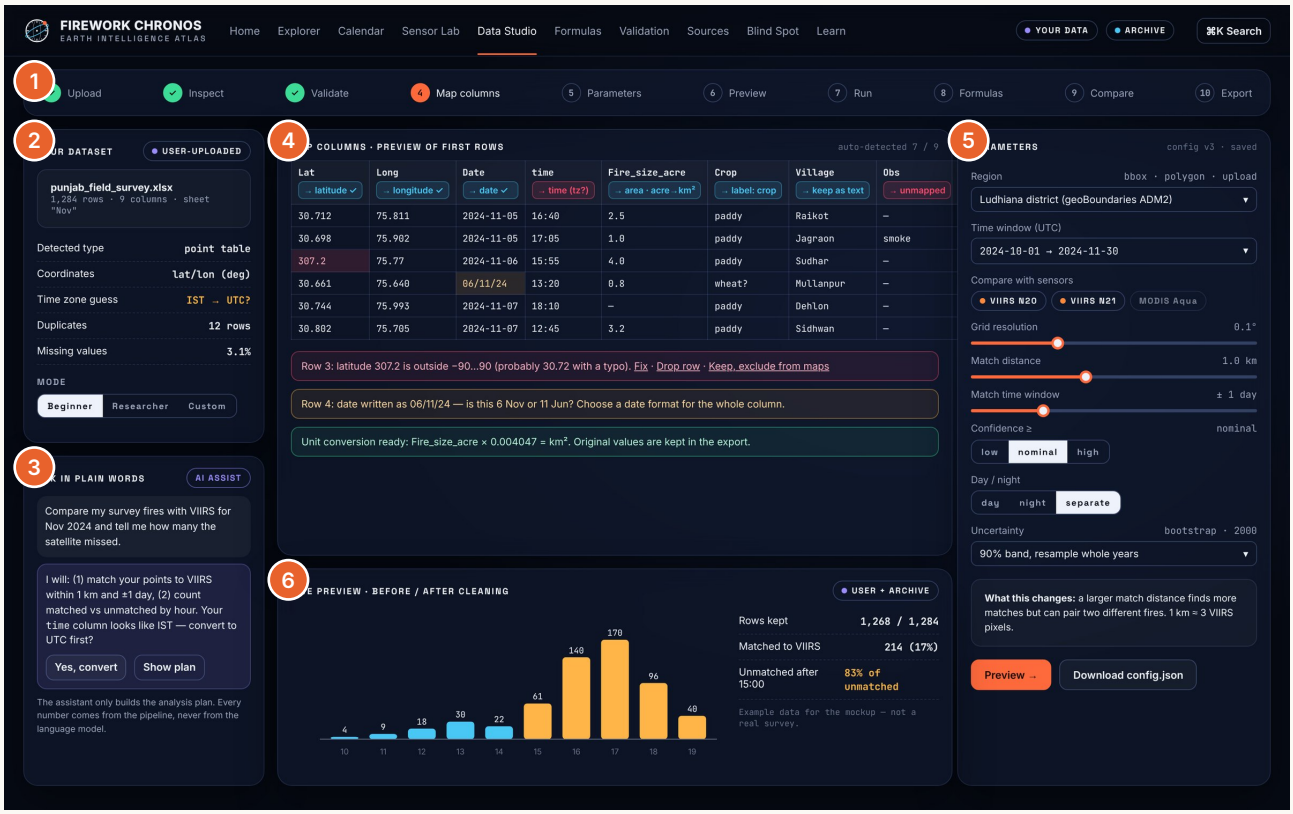
- k shown with CI, never alone
- All synthetic numbers labelled until real run

10.6 Page E · User Data Studio (/studio)

Objective: Let anyone bring their own data (file, table, paste or plain words) and run the same transparent analysis. Audience: Researchers, field teams, students, NGOs.

The User Data Studio lets anyone bring their own data and check it against the satellite record. A researcher may have a field survey of burnt farms, an NGO may have a list of complaints, a student may have a spreadsheet from a news report. The Studio accepts CSV, Excel, JSON, GeoJSON, KML, zipped shapefiles, a pasted table, or even rows typed in a chat box.

After upload, the Studio guesses which columns hold latitude, longitude, date and value, and shows them in an editable sheet so the user can correct mistakes. A validation step flags impossible coordinates, bad dates and duplicates, and converts times to UTC. The user then sets the parameters — cell size, date window, sensors, confidence, day or night — and runs a comparison. The result is a match table and a map, exported with a lineage note and tagged USER so it is never mixed into the public calendar.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. 10-step workflow: Upload → … → Export
2. Dataset card: type, coordinates, time-zone guess, duplicates, missing values; Beginner/Researcher/-Custom mode
3. Plain-words assistant: builds a plan only; numbers always from the pipeline
4. Column mapping with live validation (impossible latitude, ambiguous date, unit conversion)
5. Parameter panel with consequences explained
6. Live preview before/after cleaning

Components

- Stepper
- File inspector
- Spreadsheet grid (editable)
- Mapping dropdowns
- Validation list
- Parameter form
- Chat panel
- Preview chart
- Config download

Data

- Uploaded file (CSV, TSV, XLSX, JSON, GeoJSON)
- Schema guess
- Validation report
- Archive detections for comparison
- config.json

Interactions

- Every parameter change re-runs preview in a Web Worker (debounced 400 ms)
- Errors link to the exact row
- Before/after toggle

States

- Unsupported file: explain accepted types
- Schema unknown: manual mapping mode
- Too large (> 50 MB): suggest chunking or API import
- Server busy: job queued with position

Frontend notes

- New route; spreadsheet grid component (e.g. a lightweight data-grid) and Web Worker for parsing
- Never send files to the LLM; send only the schema summary

Backend

- POST /api/uploads (virus/size checks) → upload_id
- POST /api/jobs {upload_id, config} → job_id
- GET /api/jobs/{id} → status, result, warnings
- Files stored with TTL; user data tagged USER-UPLOADED

Limits

- Arbitrary data cannot always produce valid science; the studio says why
- Matching distance/time choices change results — shown side by side

Done when (acceptance)

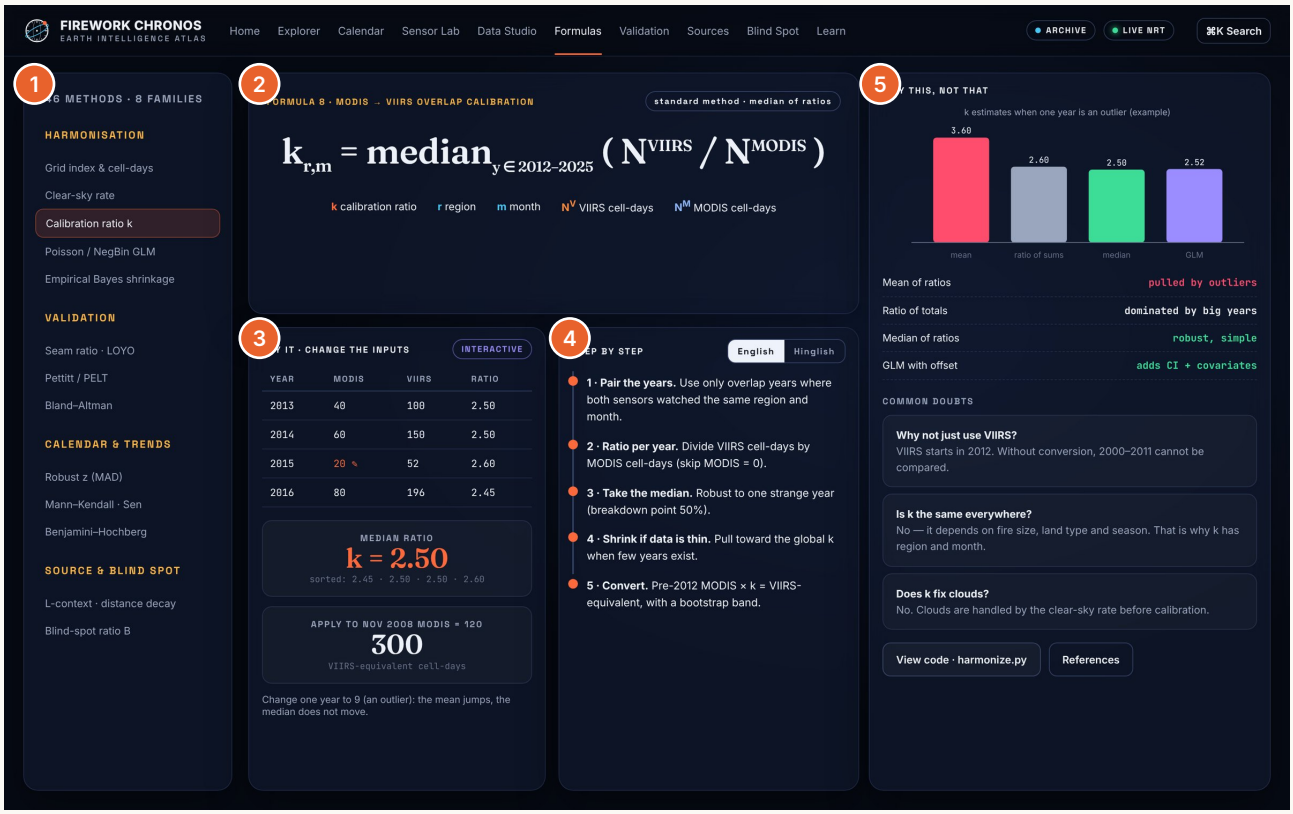
- Impossible values are always caught (lat $\notin [-90,90]$, lon $\notin [-180,180]$, future dates)
- Same config + same data $\rightarrow$ same result (reproducible)
- LLM output never contains a number not present in the job result

10.7 Page F · Scientific Formula Explorer (/formulas/:id)

Objective: Teach every formula with inputs you can change, step-by-step reasoning and honest comparisons. Audience: Students, reviewers, curious public.

The Formula Lab turns the mathematics of Part V into interactive cards. Each card shows one formula with its symbols, a short derivation, the data it uses, a live example with sliders, and a link to the code. It exists because judges reward scientific validity, and validity is easier to believe when people can change an input and watch the output move.

The page is also a teaching tool. Next to each English explanation there is a Hinglish note in plain words, and common doubts are answered under the card. Formulas that are tested carry a TESTED tag, ideas that are not yet built carry PROPOSED, and anything that uses synthetic numbers is clearly marked.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. 46 methods in 8 families
2. Formula card with symbol legend
3. Try-it table: edit inputs, see k and the converted value
4. Step-by-step in English or Hinglish
5. Why this, not that: alternatives compared, common doubts, link to code

Components

- Method index
- Formula renderer (KaTeX)
- Editable mini table
- Stepper
- Comparison chart
- Doubts accordion

Data

- Formula metadata (this Atlas, Part V)
- Small example datasets
- Links to code and references

Interactions

- Edit a value → result animates (number count-up 400 ms)
- Language toggle keeps scroll position

States

- Invalid input (e.g. MODIS = 0): explain the skip rule

Frontend notes

- KaTeX for math; content from a JSON file generated from this Atlas
- Shareable URL per formula

Backend

- Static JSON; no server needed

Limits

- Toy examples are teaching aids, not results

Done when (acceptance)

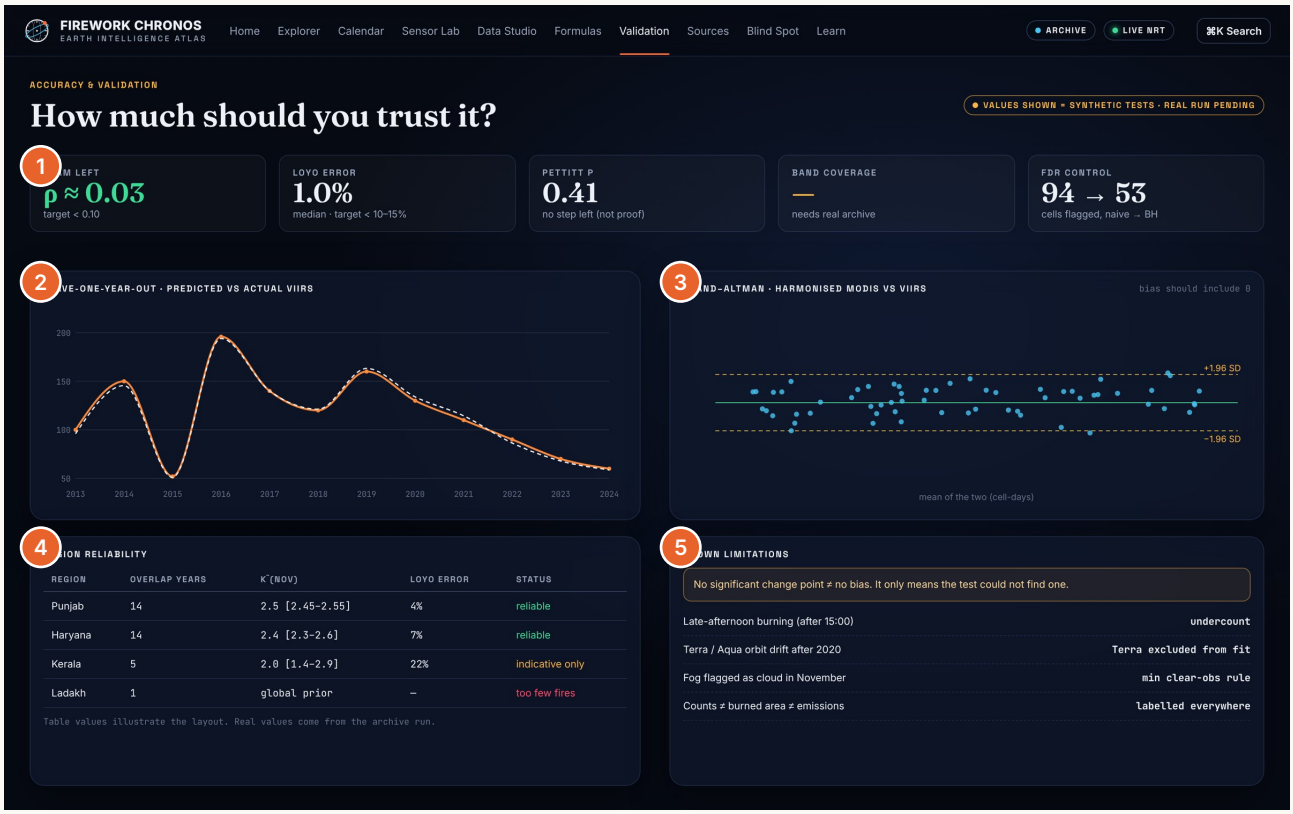
- Every formula page shows: problem, intuition, definition, example, limits, code link

10.8 Page G · Accuracy & Validation Dashboard (/validation)

Objective: Show how far each result can be trusted, where it fails, and what is still unproven. Audience: Technical reviewers, judges, scientists.

The Validation page shows how much the harmonised record can be trusted, region by region. It reports the leave-one-year-out error (how well MODIS predicts a hidden VIIRS year), the 2012 seam test before and after harmonisation, the Bland–Altman bias and limits of agreement, the model choice between a simple ratio and a power law, and whether the 90% bands really contain the truth about 90% of the time.

Pass rules are written on the page before any results are shown, so the team cannot move the goalposts. Regions that fail are not hidden: they are labelled 'indicative only'. This page is where a sceptical judge should end up, and it is designed to answer the question 'how do you know?' in under a minute.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. Scorecard tiles: seam, LOYO, Pettitt, band coverage, FDR
2. Leave-one-year-out chart
3. Bland–Altman agreement
4. Region reliability table with status
5. Known limitations

Components

- KPI tiles
- Line chart
- Scatter
- Table
- Limitations list

Data

- Validation outputs per region
- Residuals
- Coverage tests

Interactions

- Click a region row → charts filter
- Hover tile → definition popover

States

- Real run pending: tiles show '—' and a SYNTHETIC banner, never fake values

Frontend notes

- Recharts; tiles with tooltips linking to Formula Explorer

Backend

- GET /api/validation/{region}

Limits

- No significant change point ≠ no bias
- Targets are goals, not achievements

Done when (acceptance)

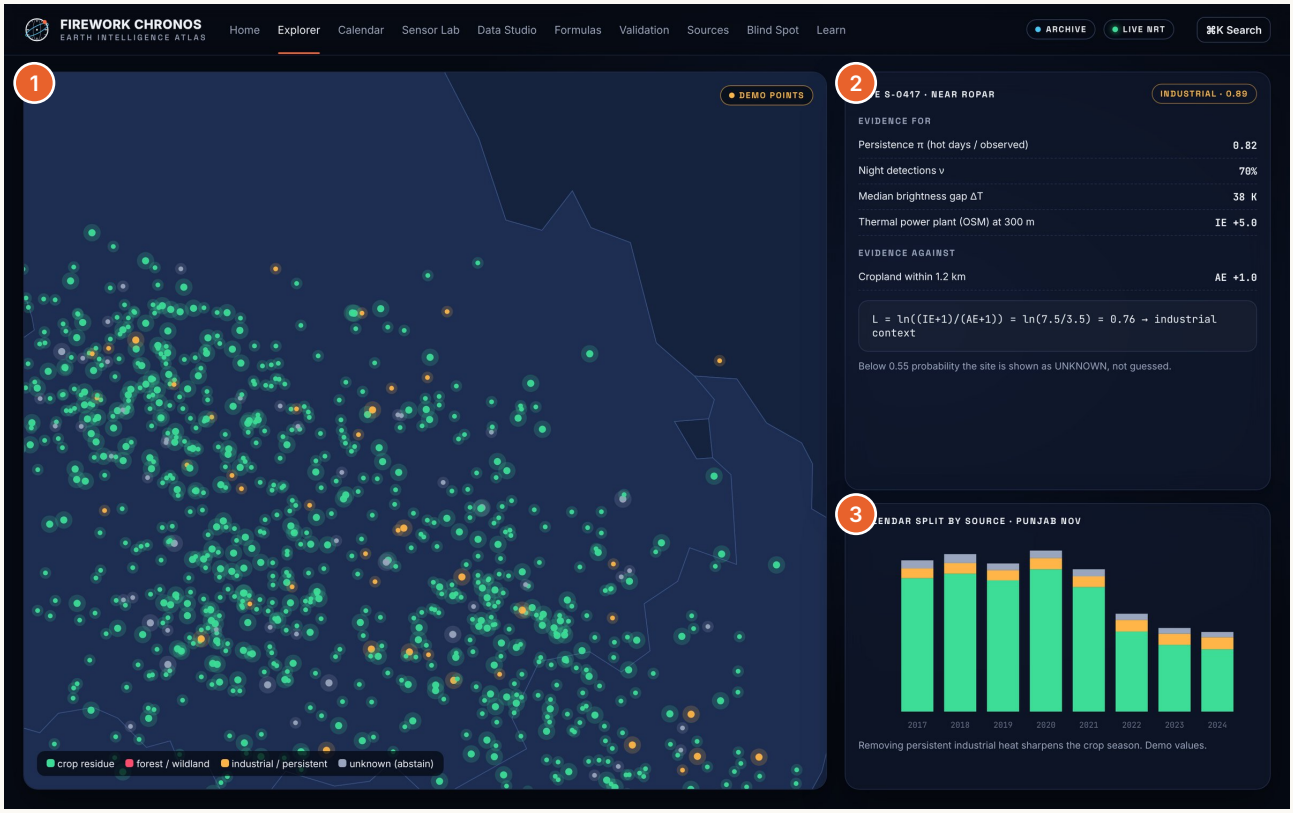
- Banner visible whenever any number is synthetic
- Each tile links to its formula

10.9 Page H · Fire Source Intelligence (/sources)

Objective: Separate crop residue, forest fire, industrial heat and unknown — with evidence for and against. Audience: Officials, researchers.

The Fire Source page tries to tell what is burning in a cell: crop residue, forest or wildland, an industrial or static source such as a brick kiln or gas flare, or unknown. It combines several clues — how many days a year the place is hot (persistence), how close it is to mapped industry versus cropland, how often it burns at night, the fire power, and land cover.

The answer is shown as a probability with an honest 'unknown' when the clues disagree. This matters because factories and flares burn all year and would otherwise inflate the agricultural season and distort the calibration between sensors.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. Map coloured by source class with legend; unknown is a valid answer
2. Site evidence card: persistence, night share, ΔT, OSM context, L value
3. Calendar split by source

Components

- Map
- Evidence card
- Stacked bars

Data

- Site features (persistence, night fraction, ΔT, IE, AE, L)
- Classifier probabilities
- OSM features

Interactions

- Click site → evidence card; 'why' expands each feature
- Filter by class

States

- Abstain (< 0.55): UNKNOWN with top-2 classes shown
- OSM missing: AE/IE shown as 'no data', not 0

Frontend notes

- Reuse SiteDossier.jsx and classify.py outputs from FIREWORK

Backend

- GET /api/sites?bbox
- GET /api/sites/{id}

Limits

- OSM coverage of kilns in India is incomplete
- Weather never used for class decisions (FIREWORK rule)

Done when (acceptance)

- Every class shows evidence for and against
- UNKNOWN never hidden

10.10 Page I · Blind Spot Observatory (/blind-spot)

Objective: Explain when satellites under-detect burning and quantify it with burned area and timing. Audience: Officials, journalists, scientists.

The Blind Spot Observatory shows what the satellites cannot see. Polar-orbiting satellites pass over a place only at fixed local times (around 10:30 and 13:30, plus night passes). If farmers burn at 4 PM, the fire may be over before the next look. The page draws a 24-hour clock with the overpass windows and the estimated burning curve, and computes the share of burning that falls inside those windows.

It also compares burned area (from MCD64A1) with the number of counted fire days. If the burnt area stays high while counts fall, the ratio B rises and the page flags that the counts are probably too low. This is our answer to the Punjab story, where reported fire counts fell sharply while burnt area fell much more slowly.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. 24-hour clock: satellite passes vs burning window
2. Counts vs burned area (reported values)
3. Blind-spot ratio B timeline with flag threshold
4. Four causes of hidden fire

Components

- Radial clock
- Bar chart

- Ratio line
- Cause cards

Data

- Burned area (MCD64A1/VNP64A1)
- Harmonised counts
- Geostationary hourly FRP if available
- Published reports (iFOREST) as context

Interactions

- Scrub the clock to see what each satellite saw

States

- No geostationary data: diurnal curve marked 'scenario, not measurement'

Frontend notes

- D3 radial clock component

Backend

- GET /api/blindspot/{region}

Limits

- Diurnal shapes are assumptions until fitted to hourly data

Done when (acceptance)

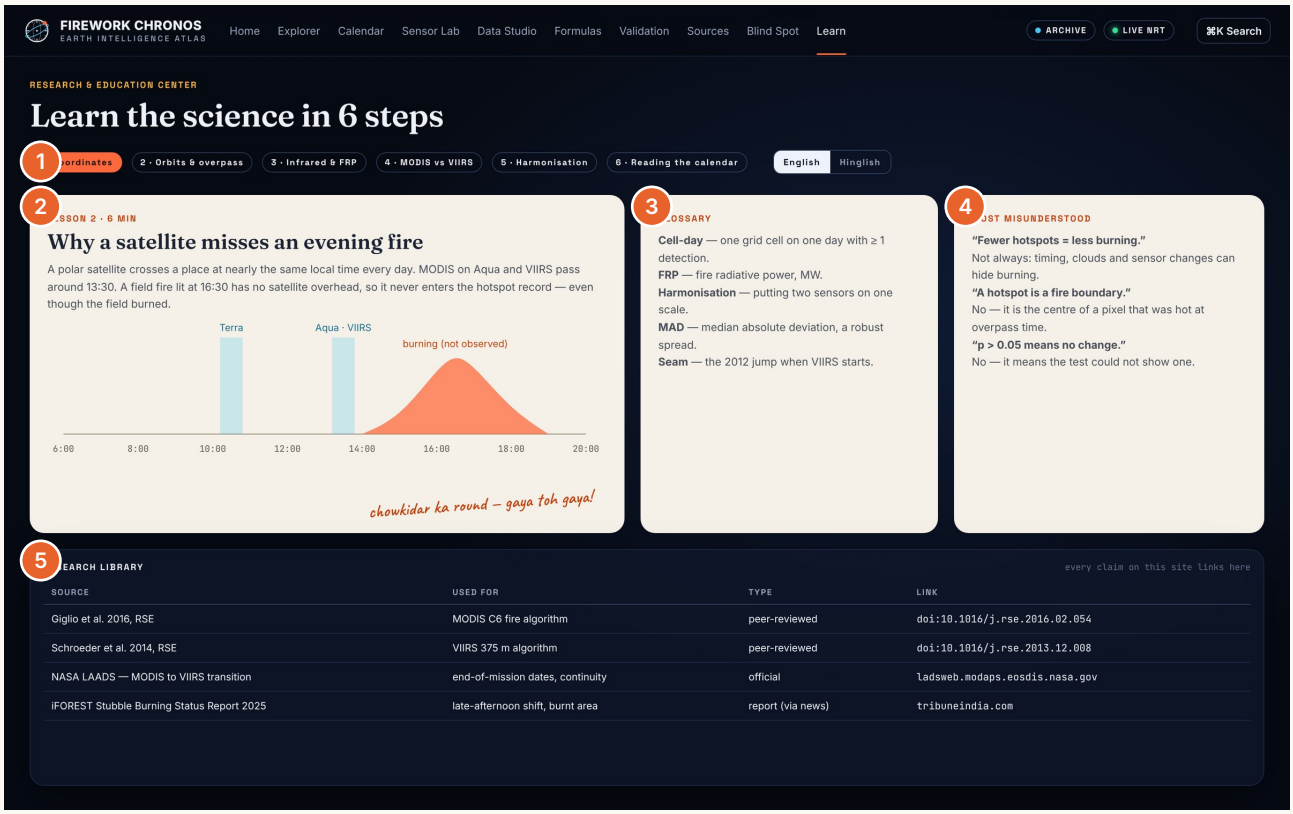
- Reported values carry source and year
- Scenario values visually different from measured

10.11 Page J · Research & Education Center (/learn)

Objective: Teach the science step by step and keep every claim linked to its source. Audience: Students, teachers, public.

The Learn page is a short course for students and newcomers. In six steps it explains how a satellite detects fire, why pixels have different sizes, why clouds hide fires, why the 2012 jump is not real, how harmonisation works, and how to read an uncertainty band. Each step has a small animation and a one-line quiz.

It reuses the same components and data as the rest of the site, so a student who finishes the course can open the Explorer and recognise every control.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. Six-lesson path + language toggle
2. Lesson card with diagram and hand note
3. Glossary
4. Most misunderstood ideas
5. Research library

Components

- Lesson path
- Lesson reader
- Glossary

- Myth cards
- Reference table

Data

- Lesson content (from Book 1 and this Atlas)
- Bibliography

Interactions

- Progress saved locally
- Each lesson ends with a 3-question check

States

- Offline: cached lessons

Frontend notes

- MDX or JSON content; paper style surfaces

Backend

- Static

Limits

- Simplified teaching diagrams are labelled as schematic

Done when (acceptance)

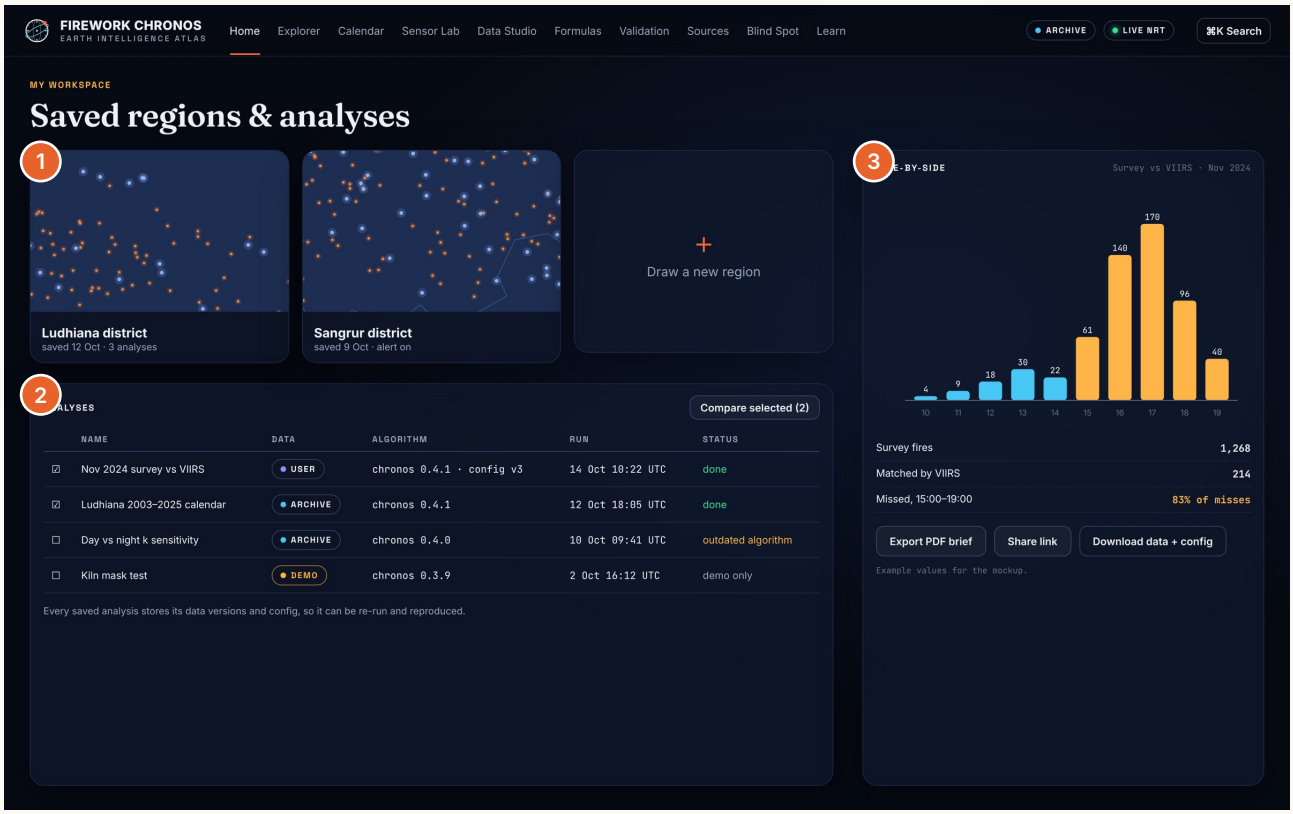
- Every lesson cites at least one source

10.12 Page K · Personalized Workspace (/me)

Objective: Save regions and analyses, compare them side by side and reproduce any result. Audience: Returning users, researchers, officials.

The Workspace is the personal area of a logged-in user. It keeps saved regions, saved views of the Explorer and Calendar, uploaded datasets from the Studio and previous comparisons. A user can return a week later and find their Punjab analysis exactly as they left it, with the parameters and data versions recorded.

Saved items are private by default. Sharing creates a read-only link that includes the full URL state, so a colleague sees the same map, the same month and the same layers.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. Saved regions
2. Analyses with data type, algorithm version, run time, status
3. Side-by-side comparison and exports

Components

- Region cards
- Analyses table
- Compare panel

Data

- User regions (GeoJSON)
- Job history with config + data versions

Interactions

- Select two analyses → compare
- Outdated algorithm → 're-run' button

States

- Not signed in: local-only workspace with warning

Frontend notes

- Simple auth (magic link) optional; local storage fallback

Backend

- /api/me/regions, /api/me/jobs

Limits

- Old results are kept but flagged when the algorithm changes

Done when (acceptance)

- Any saved result can be re-run to the same numbers

10.13 Page L · Impact & Decision Briefing (/brief/:id)

Objective: Give non-experts one printable page with the finding, its confidence and what to do. Audience: District officials, responders, journalists.

The Impact and Decision Briefing produces a one-page summary for a region and season, written for an official or a journalist rather than a scientist. It states how this season compares with past seasons, whether the change is significant, whether a blind-spot warning applies, and which districts are most unusual.

All numbers come from the calculation pipeline; a language model may only turn those numbers into plain sentences and is never allowed to invent a figure. The brief can be printed or saved as PDF and carries its data sources and generation date at the bottom.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. One-sentence headline
2. Three numbers: season, vs normal, blind-spot flag
3. What it means + suggested actions
4. Sources, algorithm version, caveats

Components

- Paper page
- KPI boxes
- Chart

- Text blocks

Data

- Calendar, season, flags for one region
- LLM-written narrative (Groq → Gemini) tagged generated_summary

Interactions

- Export PDF
- Copy share link

States

- LLM unavailable: template text, same numbers

Frontend notes

- Print CSS; paper design system

Backend

- POST /api/brief {region, period} → PDF
- narrate.py only writes words around numbers

Limits

- Must warn against over-interpreting uncertain counts

Done when (acceptance)

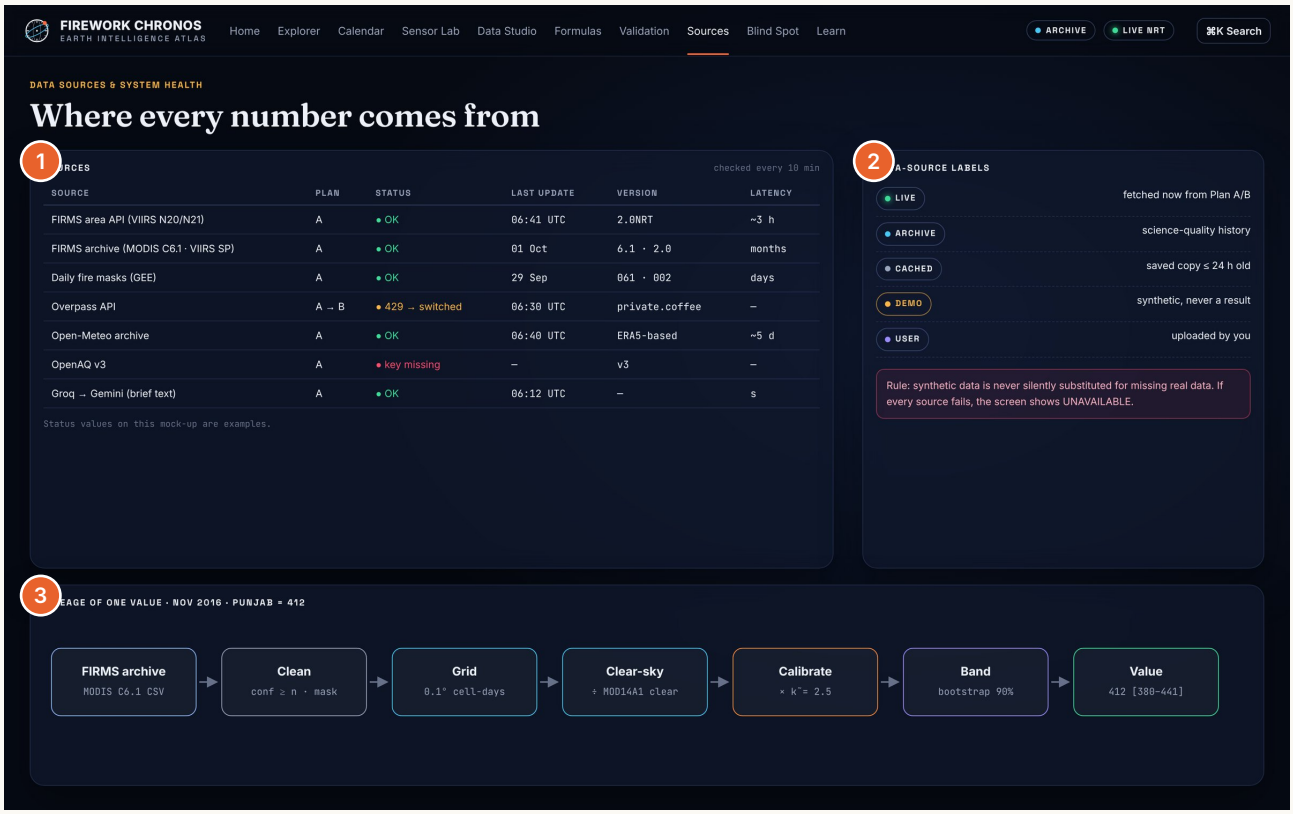
- Every number in the text exists in the data payload
- Caveat line always present

10.14 Page M · Data Sources & System Health (/sources/health)

Objective: Show where every number comes from, how fresh it is and what failed over to a backup. Audience: Developers, reviewers, power users.

The Data Sources and System Health page shows where every number comes from and whether each source is working right now. For each feed — FIRMS live, FIRMS archive, Earth Engine masks, burned area, weather, OpenStreetMap, orbit data — it lists the current status, the last successful update and which backup is in use if the main source failed.

It also shows the lineage of the current calendar: input files, code version, parameters and run time. This page is mostly for developers and judges, but it supports the honesty promise of the whole site: the user can always see whether data is live, archived, cached, demo or their own.



Hi-fi mock-up (1600 × 1000). Numbers on mock-ups are demo values. Orange numbers = marked elements listed below.

Marked elements

1. Source table: plan A/B, status, last update, version, latency
2. Data-source labels: LIVE, ARCHIVE, CACHED, DEMO, USER
3. Lineage of one value from archive to calendar

Components

- Status table
- Label legend
- Lineage diagram

Data

- health_check.py results
- Pipeline run metadata
- Dataset versions

Interactions

- Click a lineage step → shows the exact formula and code

States

- All sources failing: big UNAVAILABLE, never demo data

Frontend notes

- Simple table + SVG lineage

Backend

- GET /api/health (extend existing endpoint)
- GET /api/lineage/{value_id}

Limits

- Health shows reachability, not scientific correctness

Done when (acceptance)

- Every API response has meta.data_source
- Fallback events logged

AI

11 AI features: router, RAG and tools

11.1 AI router (backend/app/services/ai_router.py)

1. Try Groq (llama-3.3-70b-versatile or openai/gpt-oss-120b).
2. On 429/5xx/timeout try OpenRouter free model.
3. Then Gemini Flash-Lite.
4. Then GitHub Models.
5. Then template text (no AI).
6. Log provider + model in meta. Cache answers by prompt hash for 24 h.

11.2 RAG (backend/app/services/rag.py)

- Corpus: the atlas text, formula pages, dataset docs, this spec. Not raw data.
- Chunk by heading, 300–800 tokens; embed with local sentence-transformers (384-dim) or Gemini Embedding.
- Store in doc_chunk (pgvector). Retrieve top 5, answer with citations.
- Tools the model may call: get_cell(cell, month), get_calendar(region), explain_formula(id), get_health(). Numbers come only from tools.

11.3 Voice (optional)

Speech to text: Groq Whisper (whisper-large-v3-turbo, 2,000 requests/day free) with language=hi for Hindi; backup: browser Web Speech API. Text to speech: browser speechSynthesis; Google Cloud TTS free tier as an upgrade.

⚠ Free-tier warnings (Oct 2026)

Gemini free tier was cut (Flash/Flash-Lite only; prompts may be used for training). Cerebras is now a 30-day trial. Hugging Face Inference Providers needs credits. Never send user personal data to free tiers that train on prompts.

BUILD PLAN

12 Milestones for Claude Code, with done-when checks

12.1 M0 Setup

- Create folders from chapter 6; add CLAUDE.md, .env.example, .gitignore
- Unzip design reference into docs/design, data kit into data_kit/, SpaceApps_code into pipeline/ and tests/
- Install graphify and run /graphify . ; commit GRAPH_REPORT.md
- Remove seed/evaluation_metrics.json from UI; update README

Done when

pytest passes; npm run build works; docs/design previews open

12.2 M1 Exact look (all pages, fixture data)

- Copy atlas.css + fonts; build layout, top bar, StatusChip, panels exactly as the mock-ups
- Build all 13 pages + mobile versions matching preview/*.html, reading fixture JSON from frontend/src/fixtures with meta.demo=true
- Home v2 hero with earth3d.js (moving satellite, sunrise), poster image fallback

Done when

Playwright screenshots of every page match screenshots/*.jpg except data; DEMO chips visible

12.3 M2 Data engine

- resolver.py from fx_engine.py with quality gates
- sources/firms.py (API + archive CSV), gee.py, overpass.py, weather.py, orbits.py
- health_check.py feeds /api/health

Done when

resolver test: primary fails → backup used and reported

12.4 M3 Pipeline

- run.py steps 1–6 for Punjab–Haryana
- Parquet on R2; cell and cell_month tables

Done when

cell_month has rows 2003–2025; lineage row written

12.5 M4 Science

- Calibration (F10–F19), harmonisation + bands (F12, F24, F25)
- Calendar stats (F27–F32, F37), validation (F23, F33–F36), flags (F42–F46)

Done when

seam ratio and Pettitt on real data; validation table filled

12.6 M5 API

- All endpoints in chapter 9 with {data, meta}
- Caching; CORS allow-list

Done when

contract tests: every endpoint returns meta.data_source

12.7 M6 Connect pages to real data

- Replace fixtures with API calls page by page: Home, Explorer, Calendar first, then the rest
- Loading, empty, error, CACHED states

Done when

screenshots still match the look; DEMO chip disappears only where data is real

12.8 M7 Studio + AI

- Upload, mapping sheet, validation, compare
- AI router, RAG, Ask helper, briefs

Done when

upload a CSV and get a match table; LLM output contains no invented numbers

12.9 M8 Ship

- Deploy frontend (Cloudflare Pages) and API (HF Spaces / Render)
- GitHub Actions nightly + tests
- Demo rehearsal + backup video

Done when

fresh-browser test passes

CHECKS

13 Testing, deployment, open items

13.1 Testing

- Unit: every formula on a known answer (tests from SpaceApps_code.zip).
- Contract: every endpoint has meta.data_source and meta.demo.
- Golden file: Punjab Nov 2024 calendar snapshot.
- UI: Playwright for 13 routes.
- Performance: Explorer 60 fps at 100k cells; API p95 < 300 ms cached.

13.2 Deployment

1. Supabase project → run schema SQL → enable PostGIS and vector.
2. Cloudflare R2 bucket firework-data → upload Parquet.
3. Backend: Docker image to Hugging Face Space or Render; set secrets.
4. Frontend: Cloudflare Pages, build command npm run build, output dist, env VITE_API_URL.
5. GitHub Actions: nightly.yml runs pipeline/run.py and health_check.py.

13.3 Open items (decide or provide)

- OPEN: 2026 rules on code written before 14 Nov.
- OPEN: final regions beyond Punjab–Haryana.
- OPEN: geostationary FRP access for India (LSA SAF IODC, MOSDAC) — unverified.
- OPEN: friend's globe project link (not available).
- Re-check every free-tier limit the week before the hackathon.

13.4 Related documents

- FIREWORK_CHRONOS_Atlas_A4_Report.pdf — full research, design, 46 formula pages.
- FIREWORK_Technical_Approach_Free_Stack.pdf — free stack, databases, ORMs, AI.
- FIREWORK_APIs_Keys_Datasets.pdf — every API response field by field.
- Strategy, accuracy upgrade and API keys notes in the NASA project.